Roma Tre

Information Visualization

Perception and Congnitive Issues

M. Patrignani

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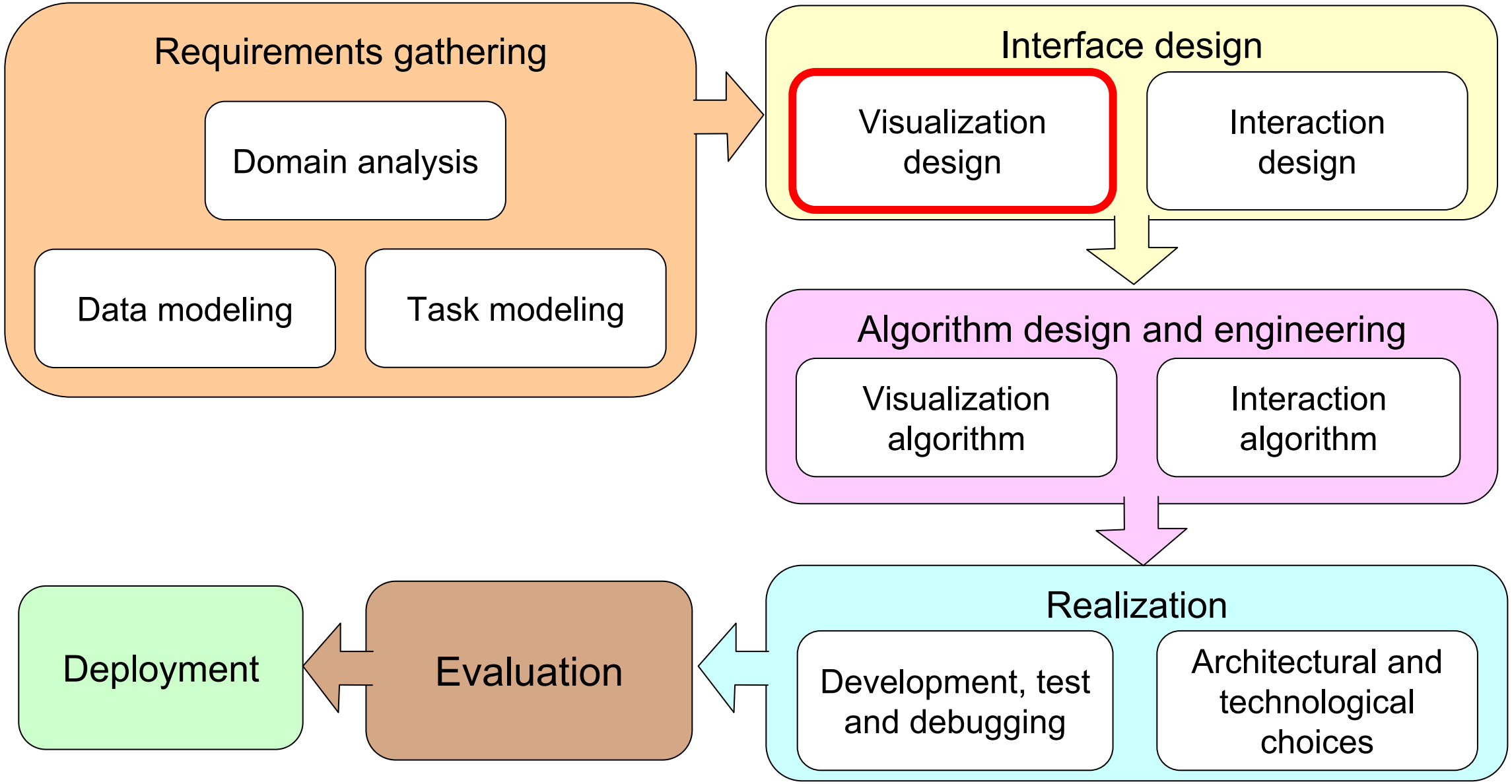
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Summary

- Our perception abilities
  - gestalt laws
  - preattentive processing
- Interferences in perception
  - erroneous perceptions of lengths and areas
  - Stevens' power law
  - Weber's law
- Mapping variables to graphics

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Design pipeline



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Our perception abilities

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Gestalt laws

- The name comes from the gestalt school of psychology (Germany, 1912)
  - “gestalt” means “pattern” in German
  - main researchers were Max Westheimer, Kurt Koffka, and Wolfgang Kohler
- They produced a set of gestalt laws of pattern perception
  - these are robust rules that describe the way we see patterns in visual displays
  - they easily translate into a set of design principles for information displays

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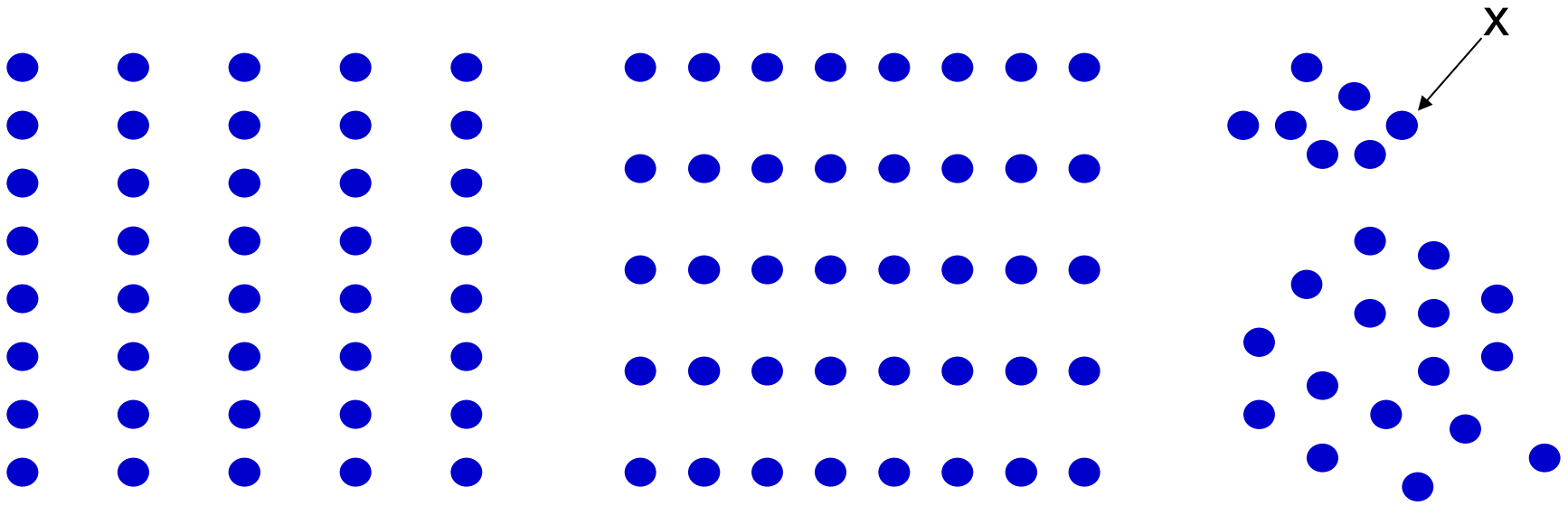
Gestalt laws

- The main gestalt laws are the following six
  - Proximity
  - Similarity
  - Connectedness
  - Continuity
  - Symmetry
  - Closure

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Proximity

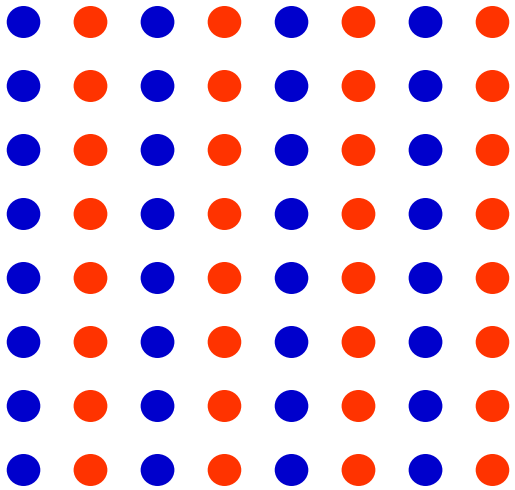
- Things that are close together are perceptually grouped together
- One of the most powerful gestalt laws
  - also called “spacial concentration”

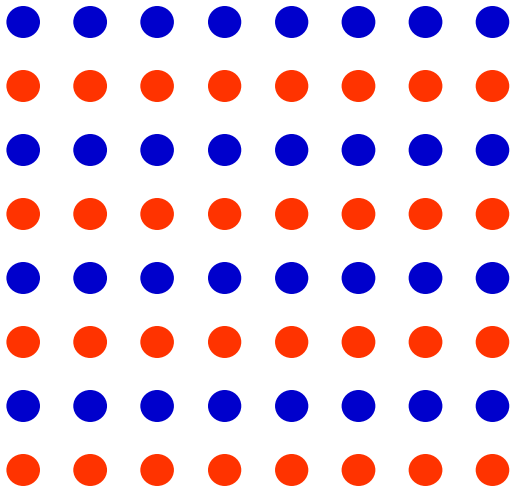


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Similarity

- Similar elements tend to be grouped together
  - in this example we perceve rows or columns because of the similar color

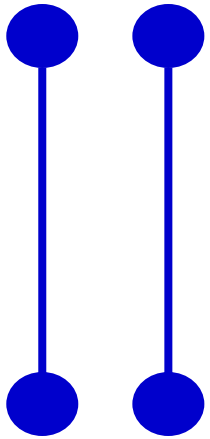




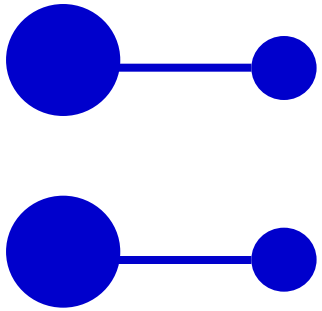
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Connectedness

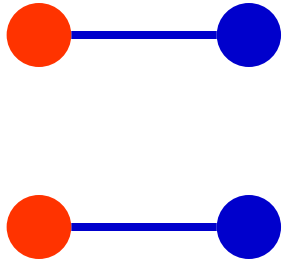
- We associate connected things together
- Stronger than proximity, color, size and shape
  - four examples



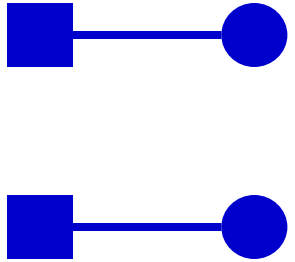
Connectedness beats Proximity



Connectedness beats Similarity (size)



Connectedness beats Similarity (color)

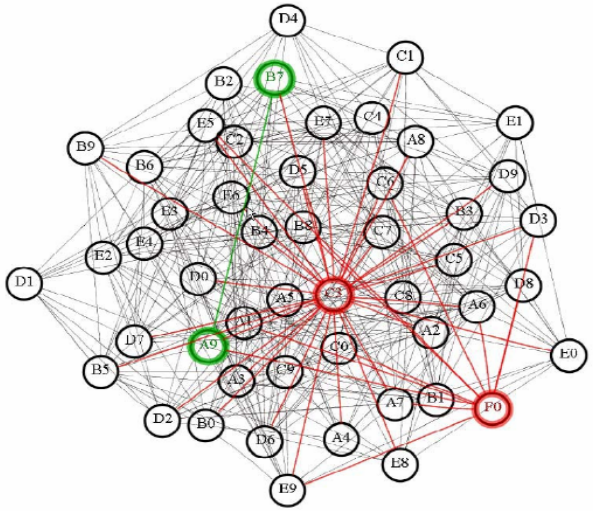


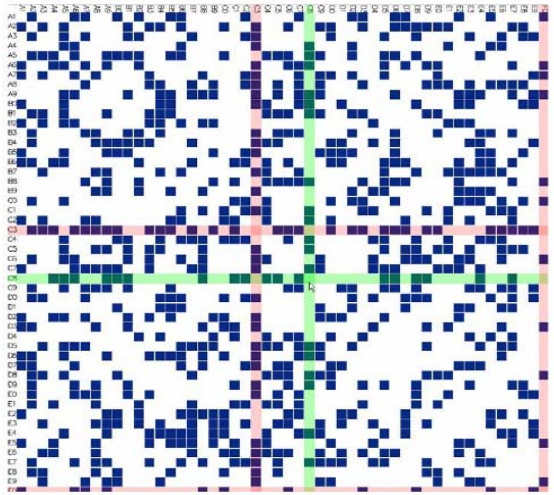
Connectedness beats Similarity (shape)

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Exploiting Connectedness

- Based on connectedness, binary relationships have a natural representation, which is called “node-link”
  - as opposed to matrix-based representations
  - confirmed as more effective when graphs are small and paths are involved [Ghoniem et al, 04]



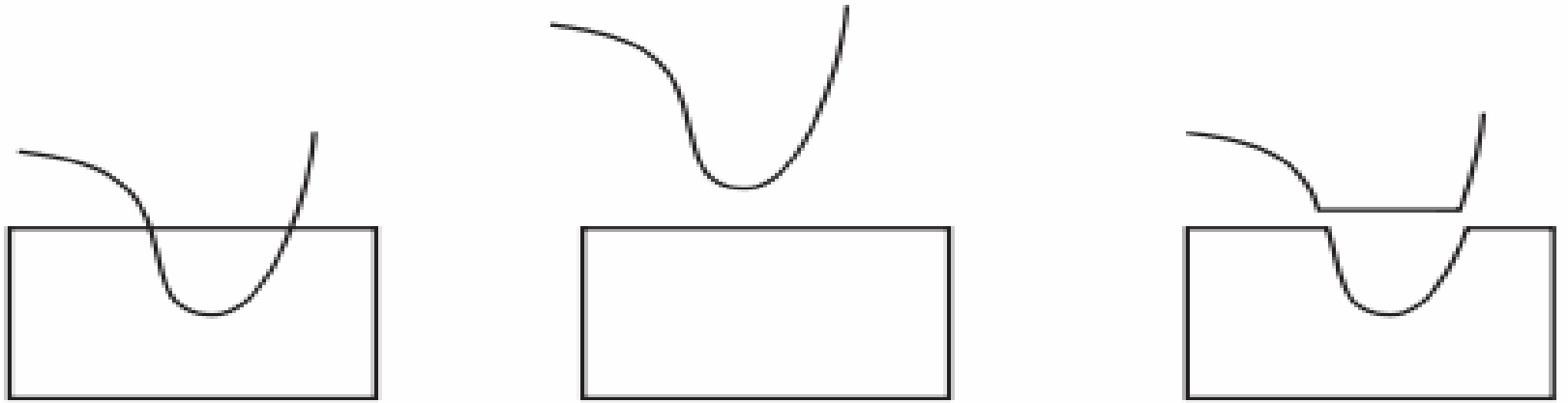


Pictures from [Ghoniem et al, 04]

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Continuity

- We are more likely to construct visual entities out of visual elements that are smooth and continuous, rather than ones that contain abrupt changes in direction
  - example from [Ware 13]

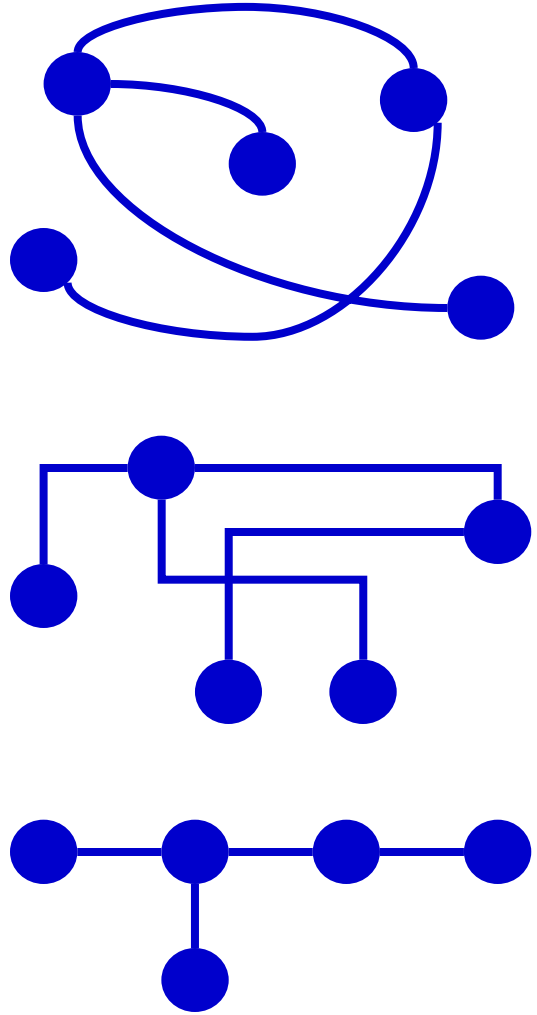


Stimulus Interpretation one Interpretation two

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Exploiting Continuity

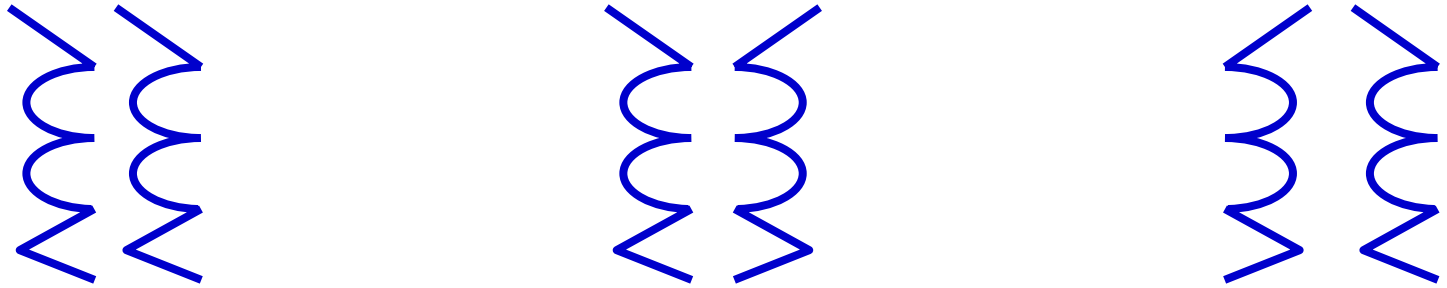
- Representing binary relationships as graphs
  - crossings are discontinuities
  - in orthogonal representations bends introduce discontinuities
- Most of the drawing algorithms try to produce representations where crossings and bends are reduced



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Symmetry

- Symmetric shapes are perceived more strongly as a visual whole
  - example from [Ware 13]

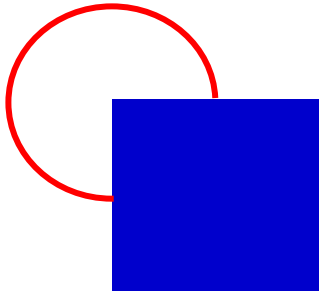


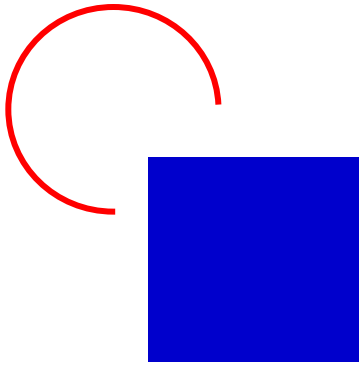
- Most effective symmetries are along the horizontal and vertical exes
- Only small symmetric patterns can be detected

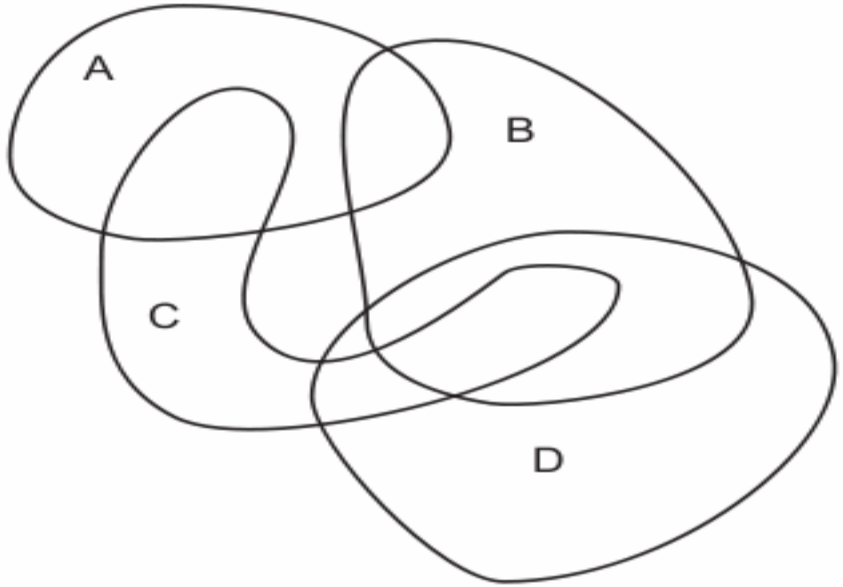
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Closure

- A closed contour tends to be seen as an object
- There is a perceptual tendency to close contours that have gaps in them







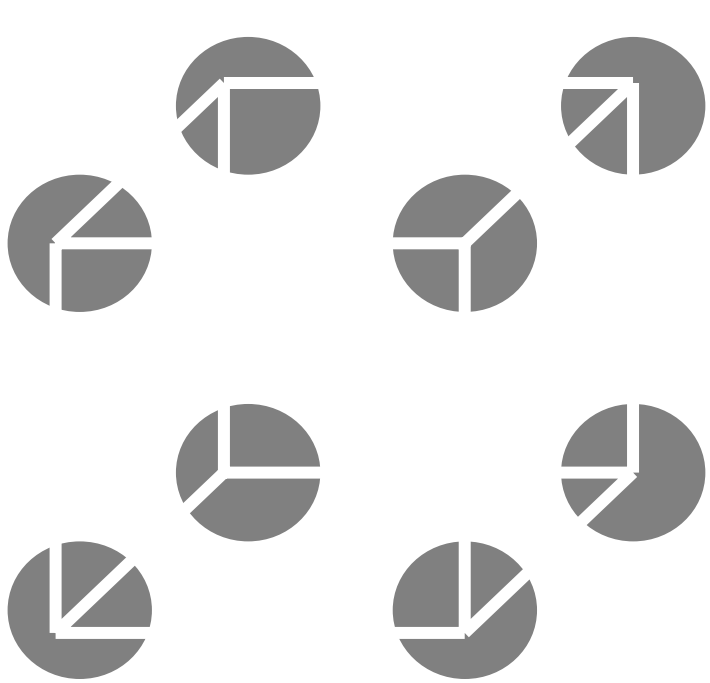
Euler diagram

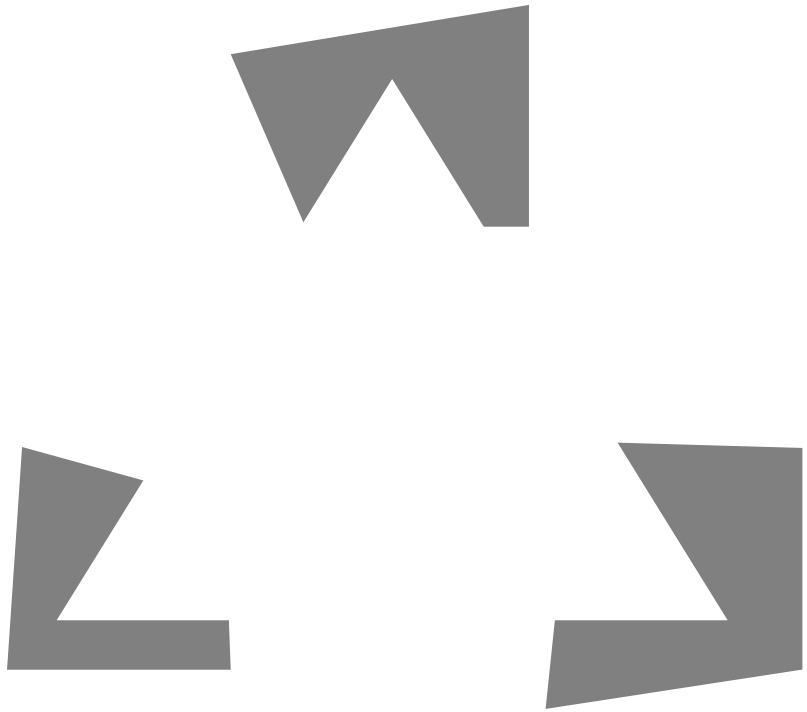
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Examples of closure

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Examples of closure





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The perception process

- Till now we considered some perception principles
- In the following we will consider the performance of perception
  - involves comparing alternative perception scenarios
  - gives an insight on the perception process

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Perception experiment

- how many 3s do you see in this table?

459290780597720987759726556651100498366 45
271074621446542070790147381097438970109 71
439070973492668478587158190486309018890 74
257470723547456661420187740728498753106 65

  - you have to scan the numbers sequentially

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Perception experiment

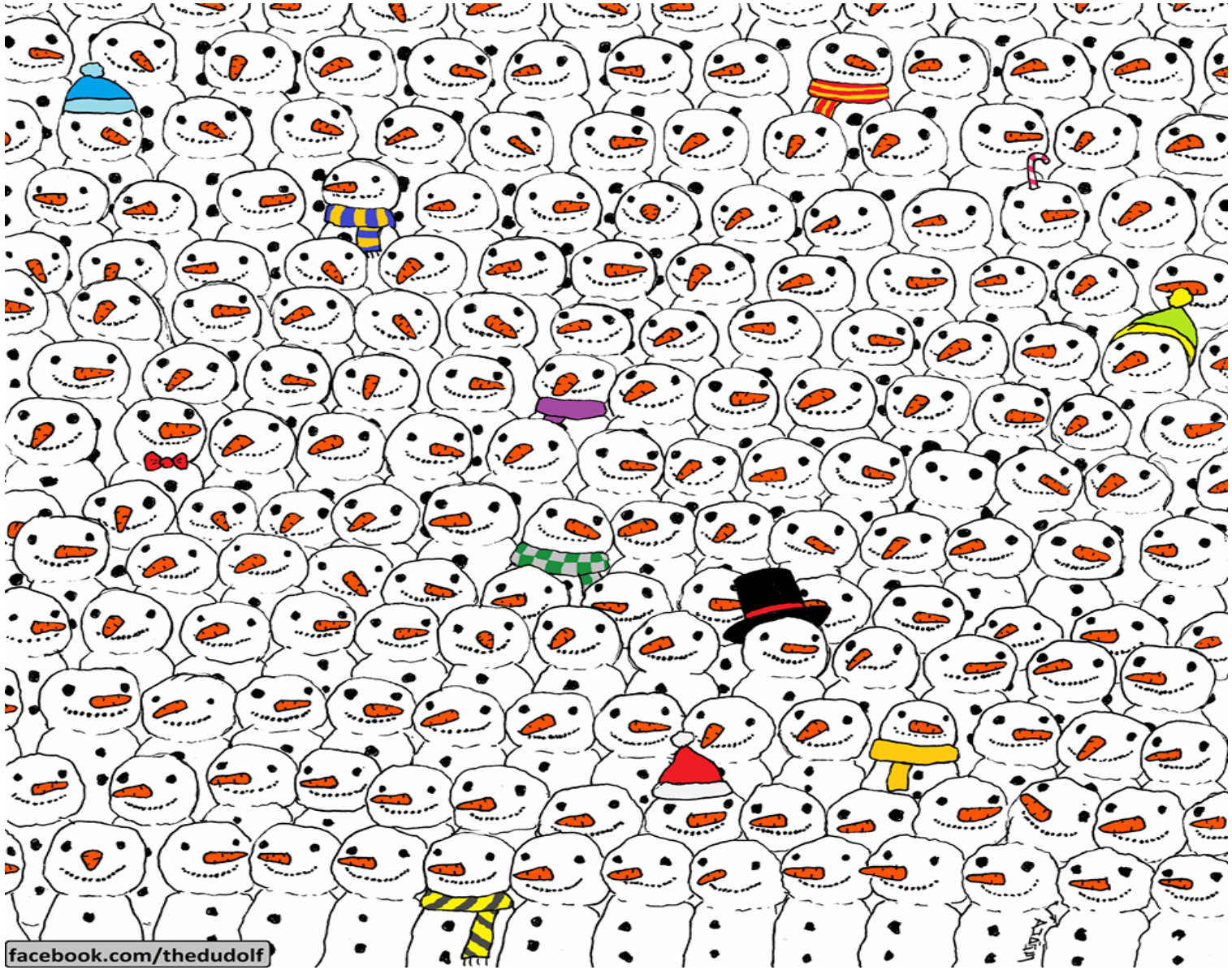
- how many 3s do you see in this table?

4592907805977209877597265566511004983 6645
2710746214465420707901473810974389701 0971
4390709734926684785871581904863090188 9074
2574707235474566614201877407284987531 0665

  - now they pop out from their surroundings
    - this example is from [Ware 13]

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Perception game: can you see the panda?



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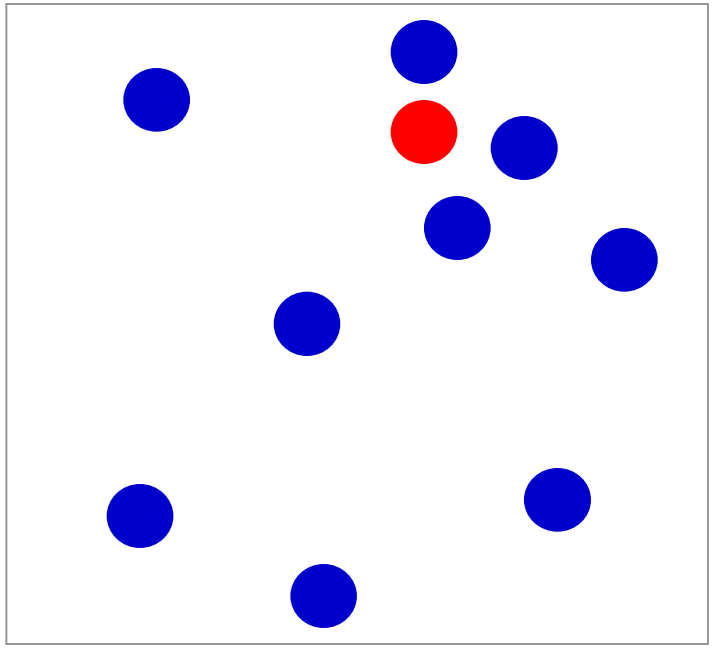
Preattentive processing

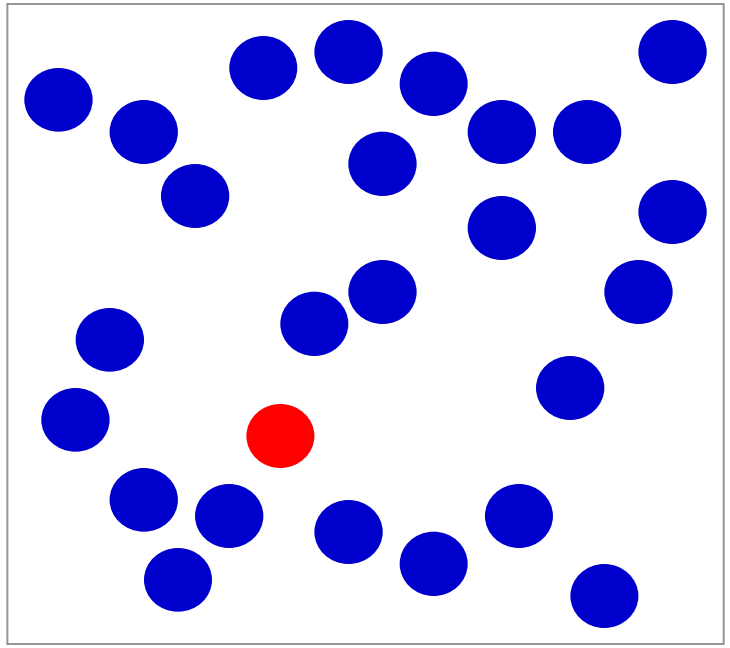
- Certain simple shapes or colors seem to pop out from their surroundings
- The theoretical mechanism underlying popout is called preattentive processing
- Preattentive processing
  - no need to focus attention
  - perceived immediately
    - less than 200/250 ms qualifies as preattentive
    - eye movements take at least 200ms
    - important for visual interfaces design
  - implies low-level parallel processing

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Visual popout of color

- A different hue allows objects to pop out from their surroundings
  - this ability is indepedent from the number of the so-called “distractors”

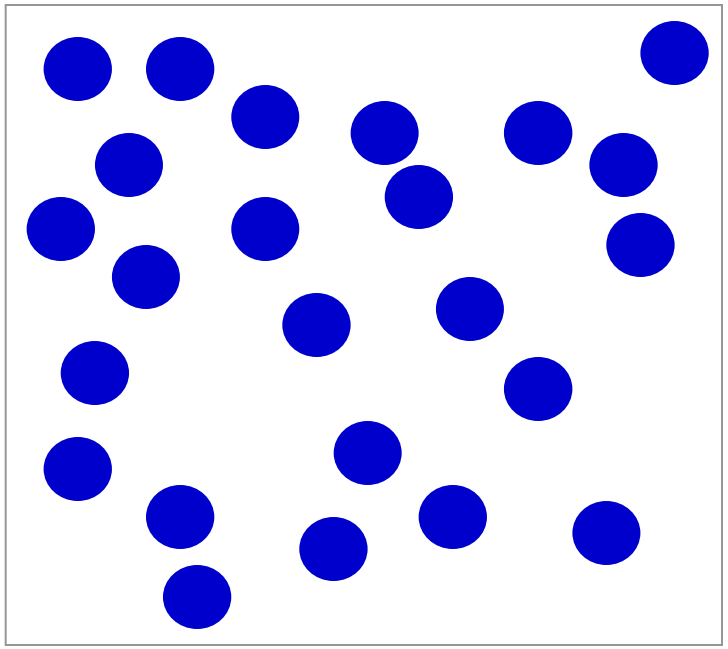


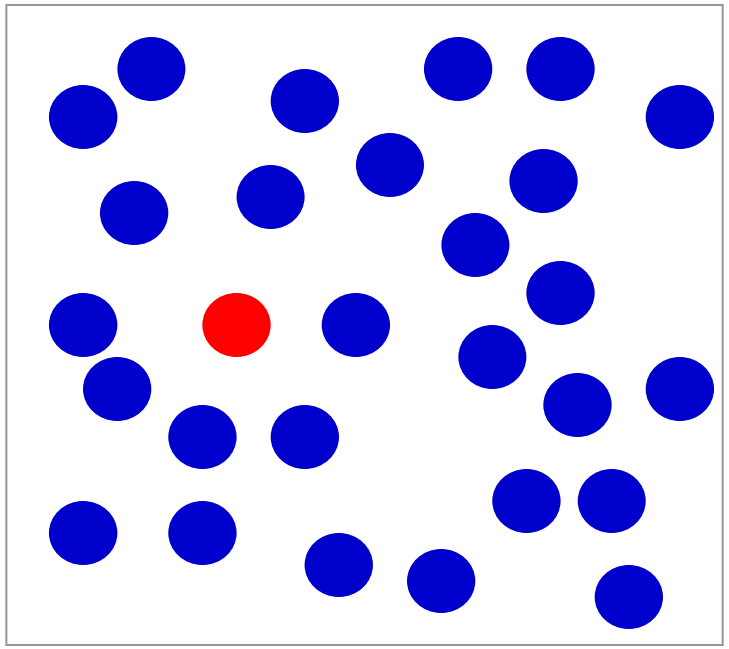


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Preattentive processing tests

- Ask to determine if a target is present among a set of other distractors
  - for example: search  among distractors 
- Measure the response time with respect the number of distractors

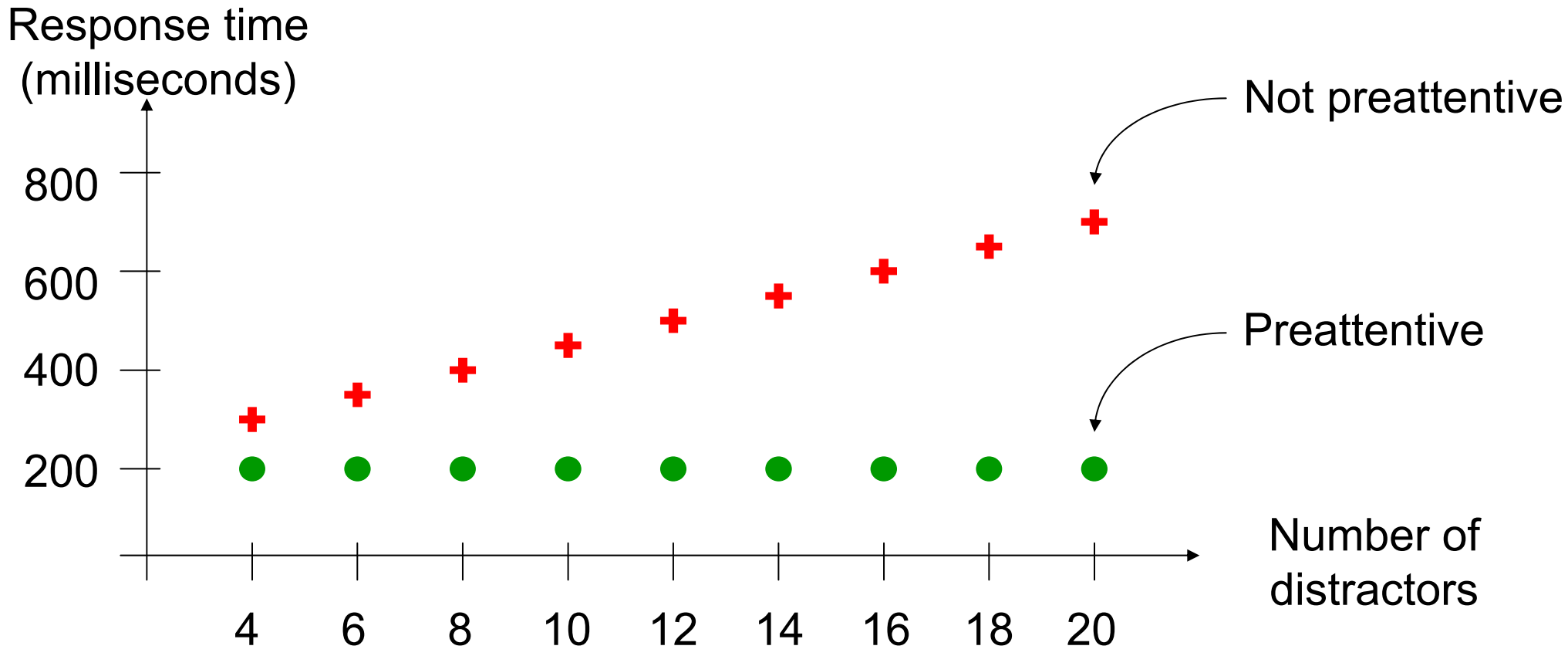




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Preattentively processed features

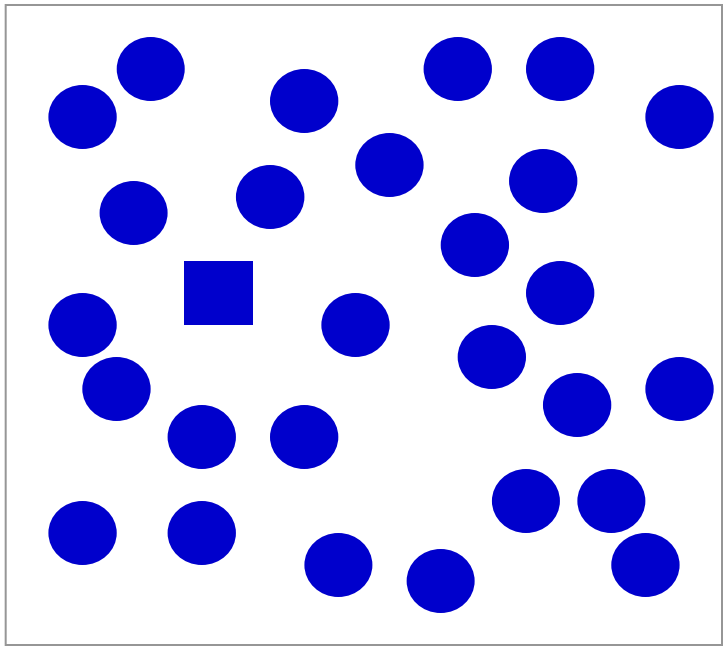
- A feature is preattentively processed if the response time is constant

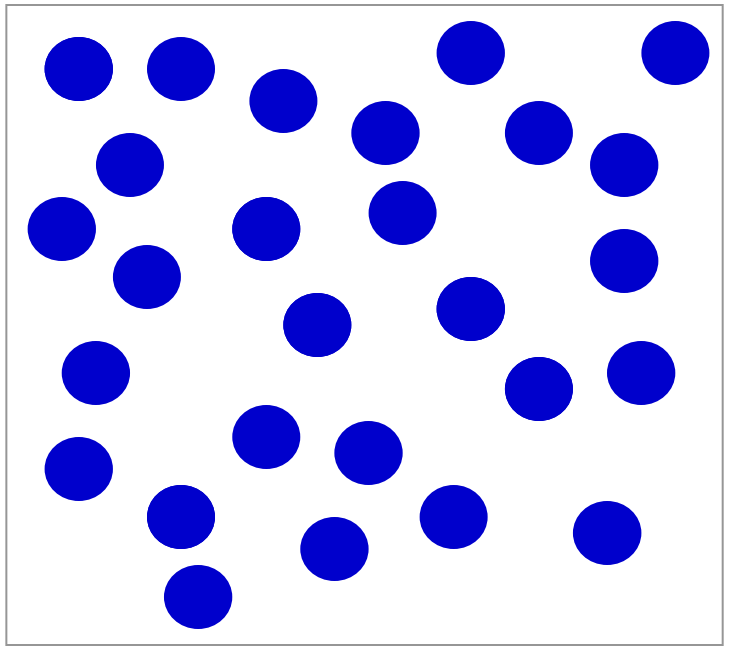


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Visual popout of shape

- Search for the square ■ among the many circles ●
  - a distinct shape also pops out





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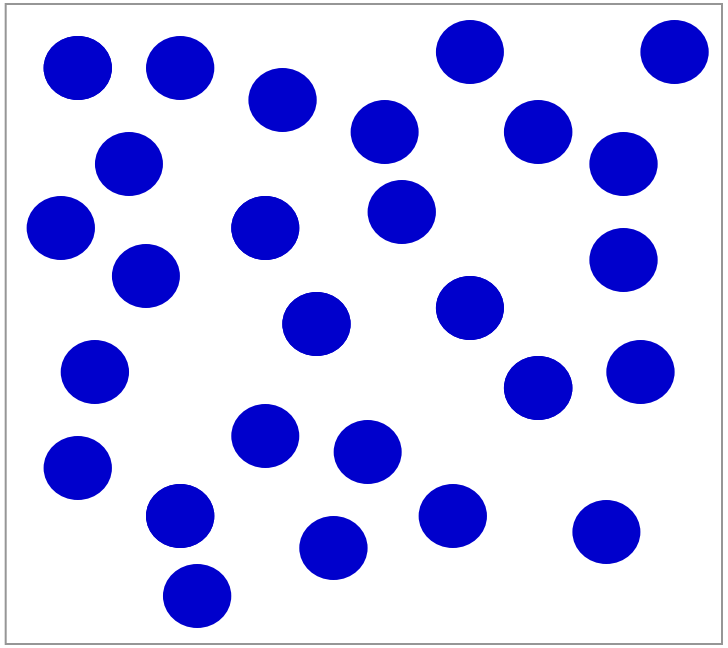
Popouts and channels

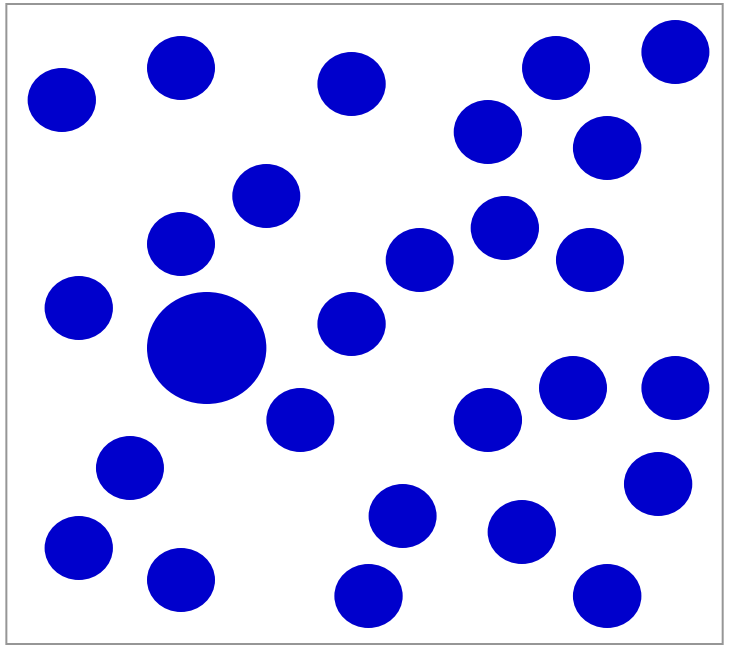
- When a graphic feature makes the target pop out, we say that it is a “channel”
- Researchers try to experimentally find out how many different channels are there
- Developers use different channels to convey different information
  - color and shape are two different channels

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Visual popout of size

- Search for the big circle ● among the many small circles •
  - a distinct size pops out

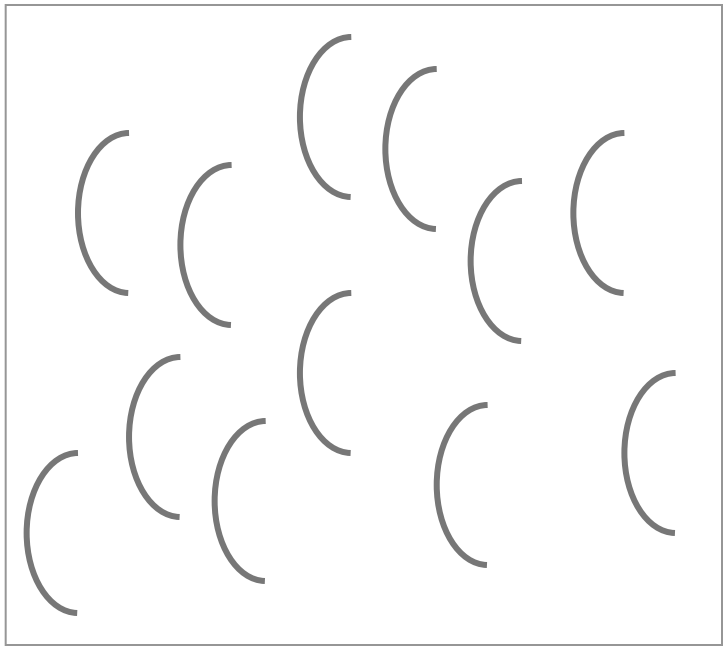


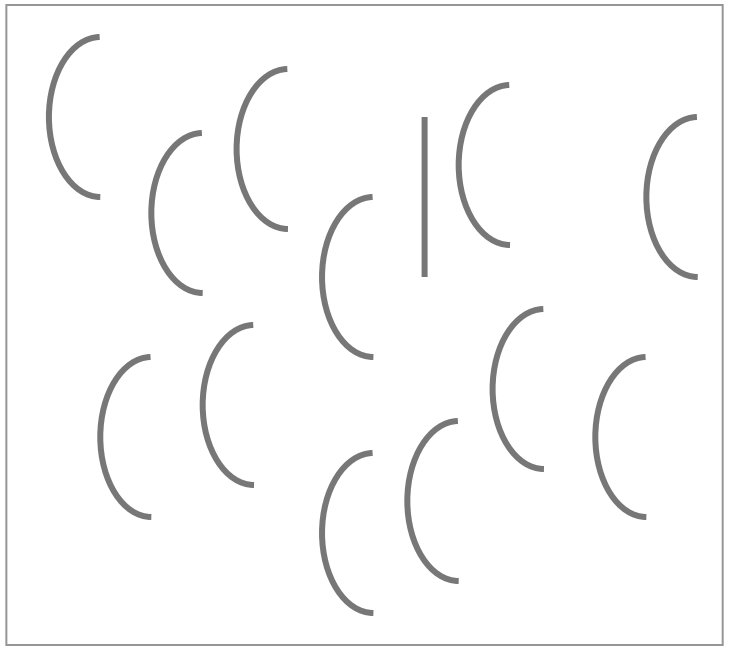


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Visual popout of curvature

- Search for the big straight-line segment | among the many curved ones (
  - curvature pops out

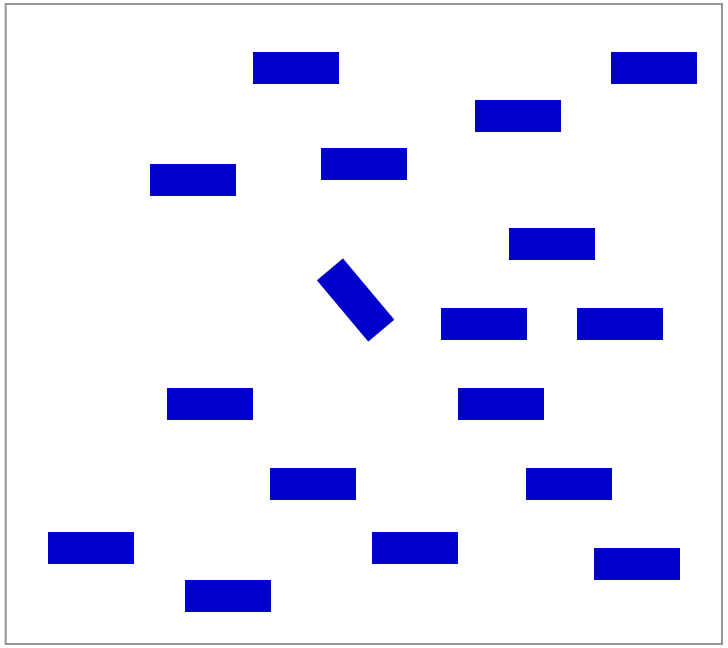


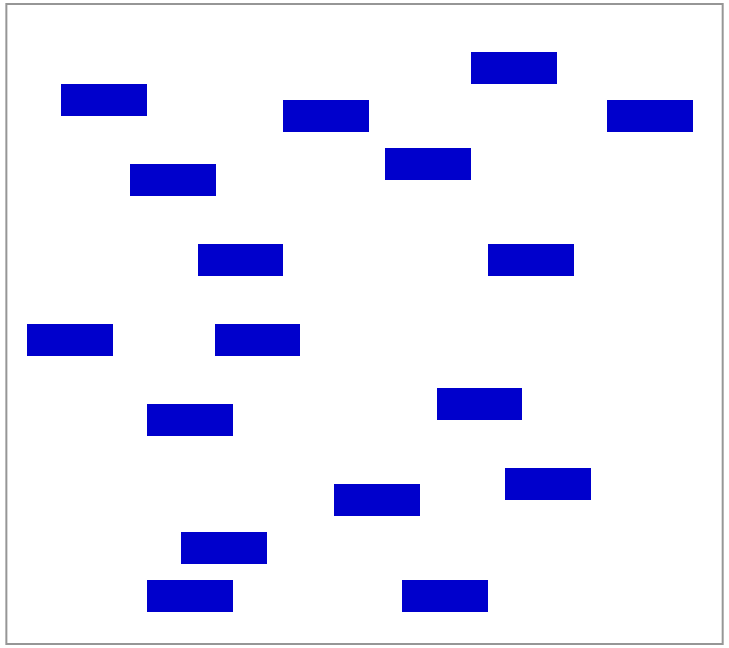


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Visual popout of orientation

- Search for the tilted rectangle ◆ among the many horizontal ones ▬
  - orientation pops out [Julész & Bergen 83]

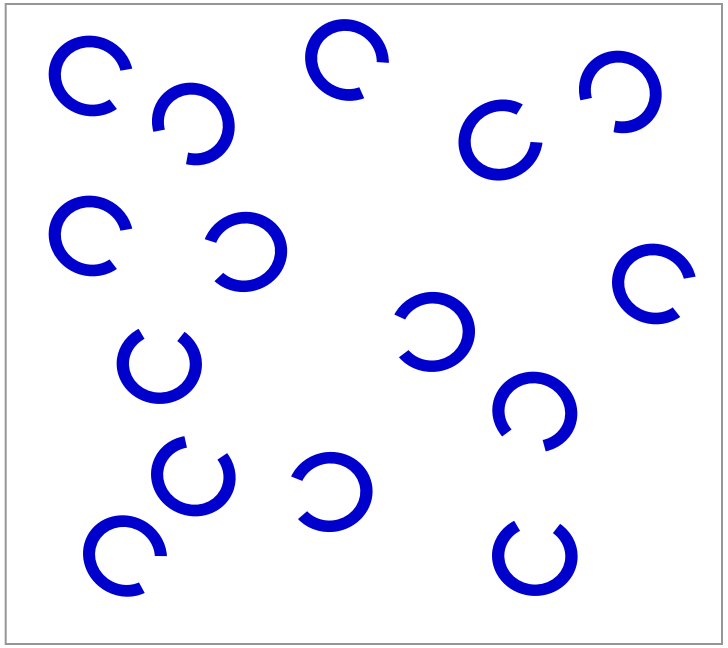


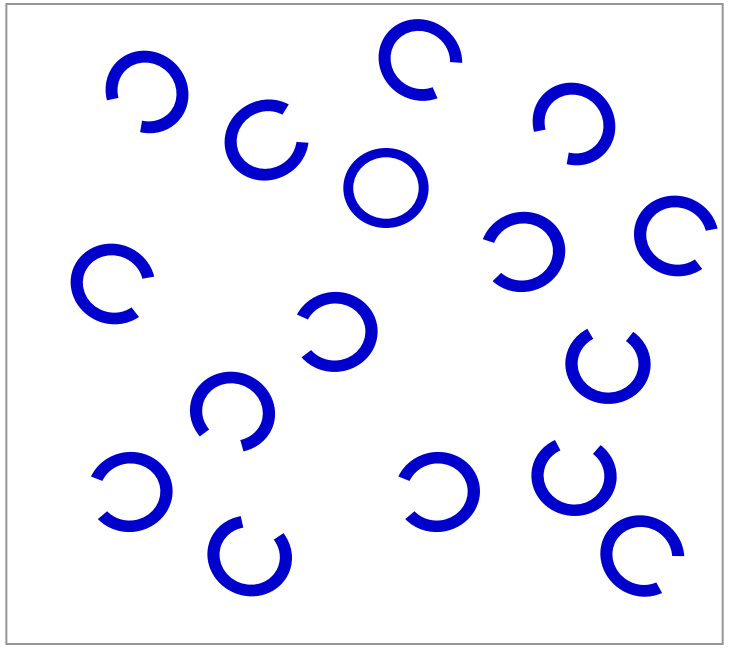


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Visual popout of closure

- Search for the closed shape ○ among the many open ones ⊂
  - closure pops out [Julész & Bergen 83]

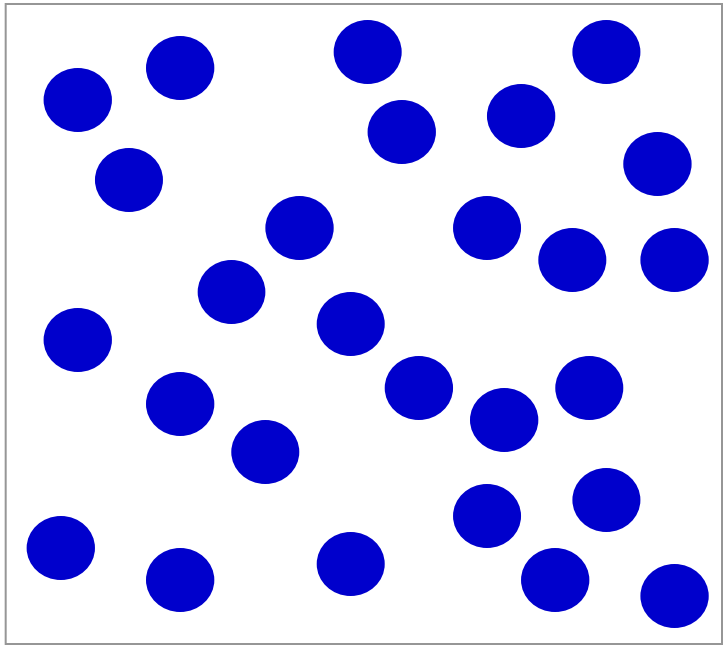


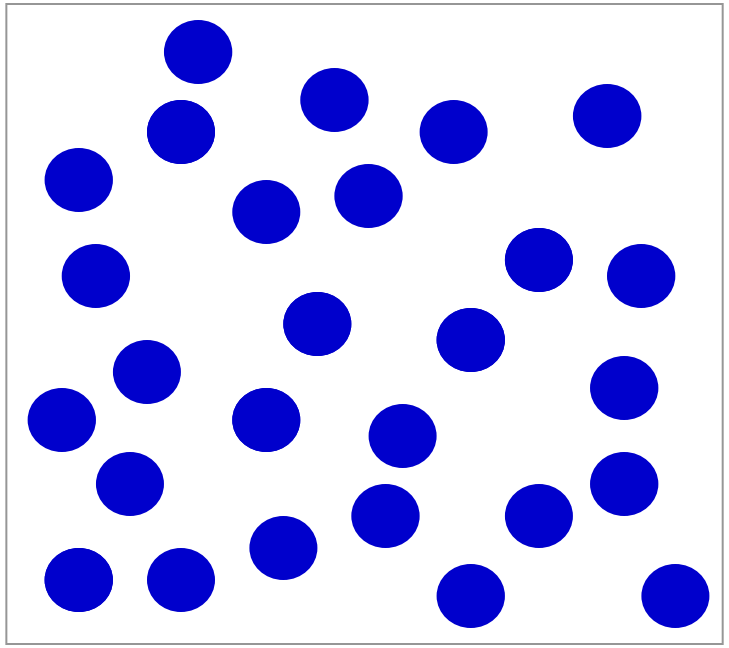


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Visual popout of motion

- Search for the moving circle ●
  - moving objects pop out





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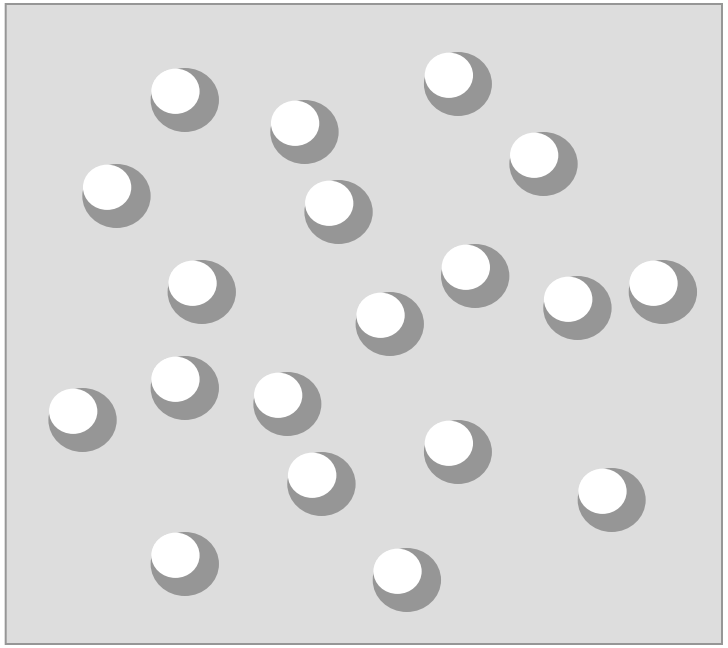
Use of motion for highlighting

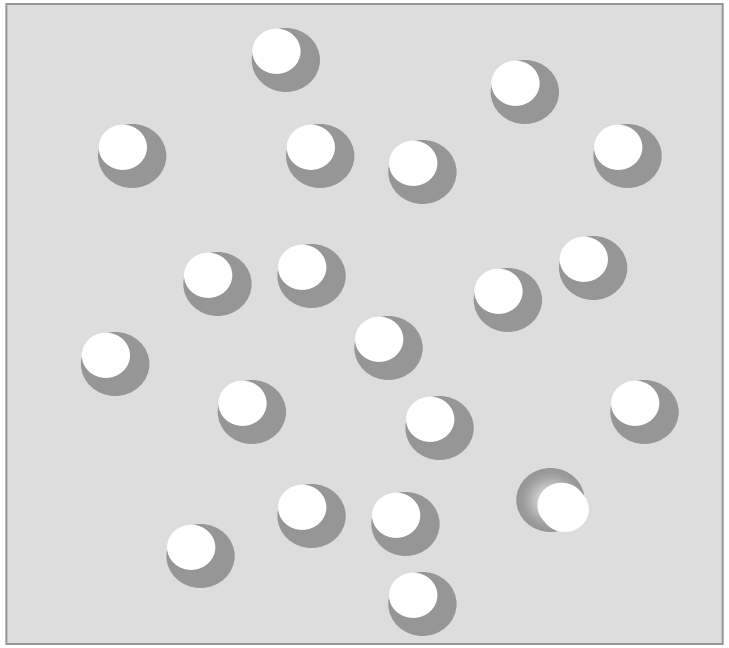
- Modern computer graphics allow to use motion for highlighting
- Motion is very effective
  - however, making things move may be too strong a cue for many applications
- Similar behaviours have an analogously strong highlighting effect
  - flicker and blinking
  - direction of motion

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Convex and concave

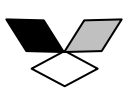
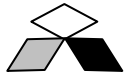
- Search for the concave among the convex
  - convex/concave objects pop out

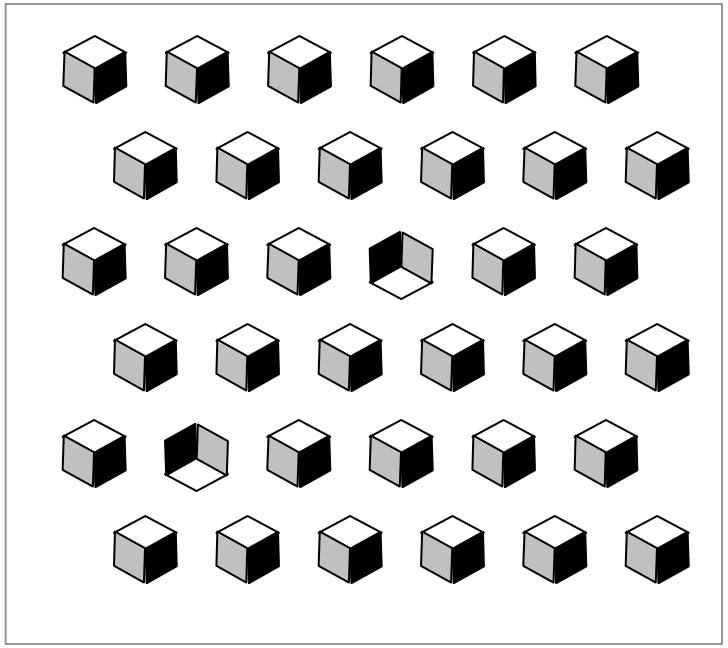


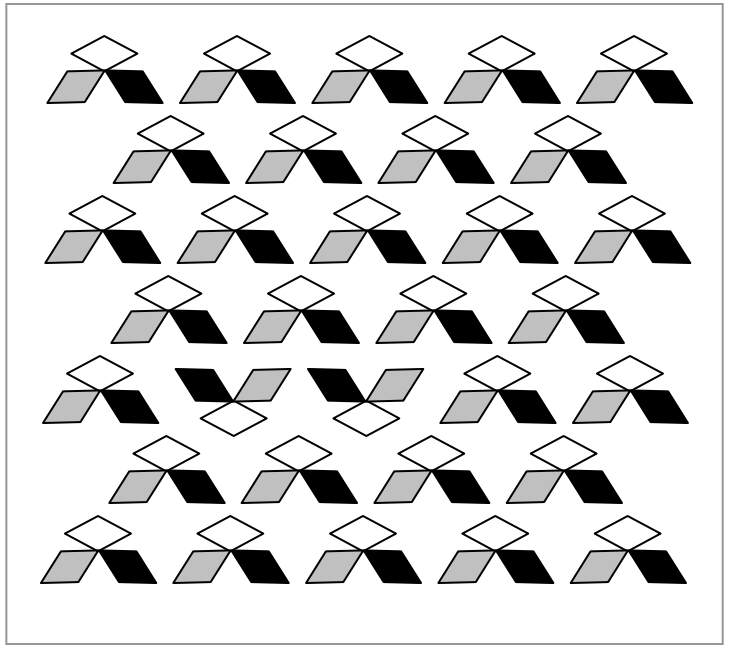


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Convex and concave

- A second example of convex/concave with  and  shapes
  -  and  shapes are ineffective





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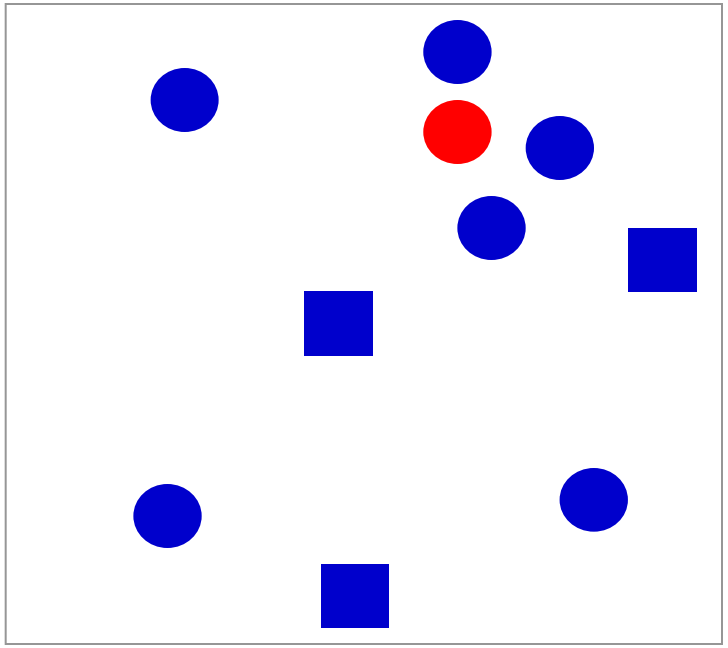
Channels considered insofar

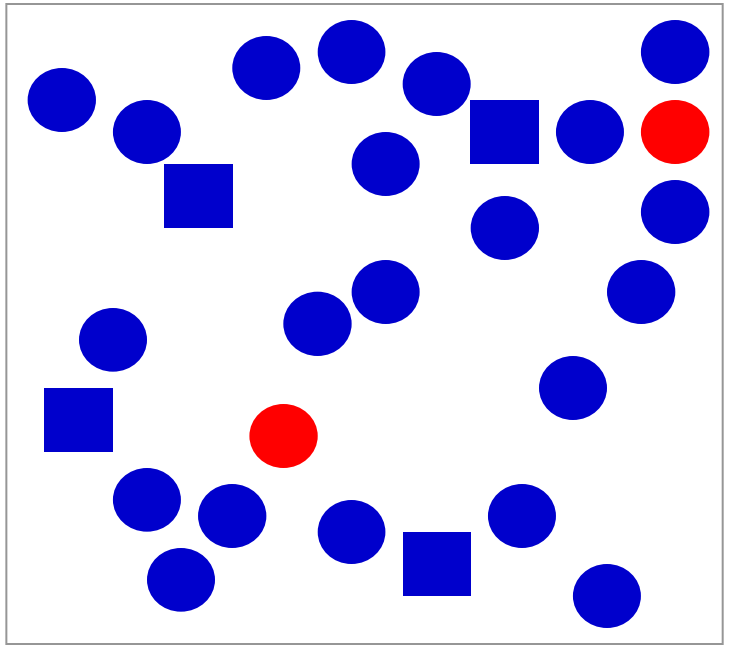
- Color
  - hue
  - density, contrast
- Shape
  - curvature, closure
- Size
  - length, width
- Line orientation
  - intersections, terminators
- Motion
  - direction and speed of motion
  - flicker, blinking
- 3D depth cues
  - convex, concave, lighting direction

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Preattentive processing is tunable

- We can tune preattentive processing on specific visual properties
  - ask your eyes to search for blue squares
  - ask your eyes to search for red circles

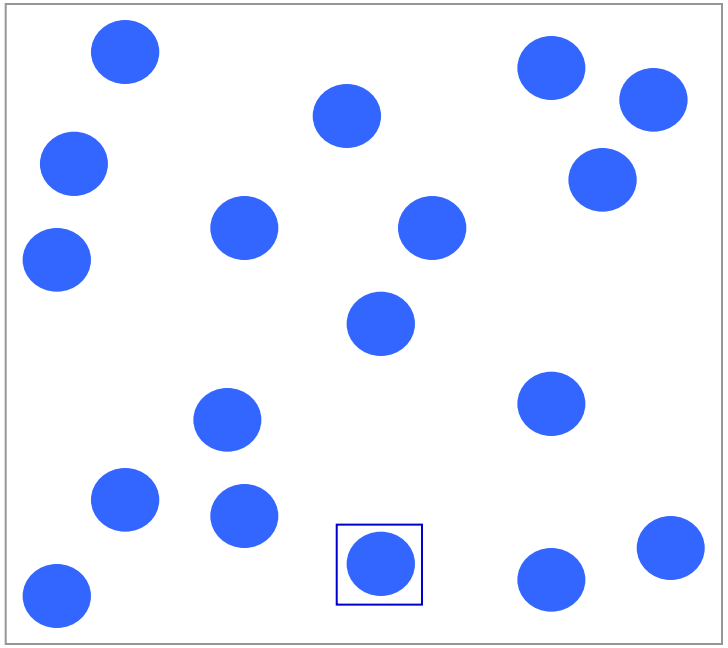


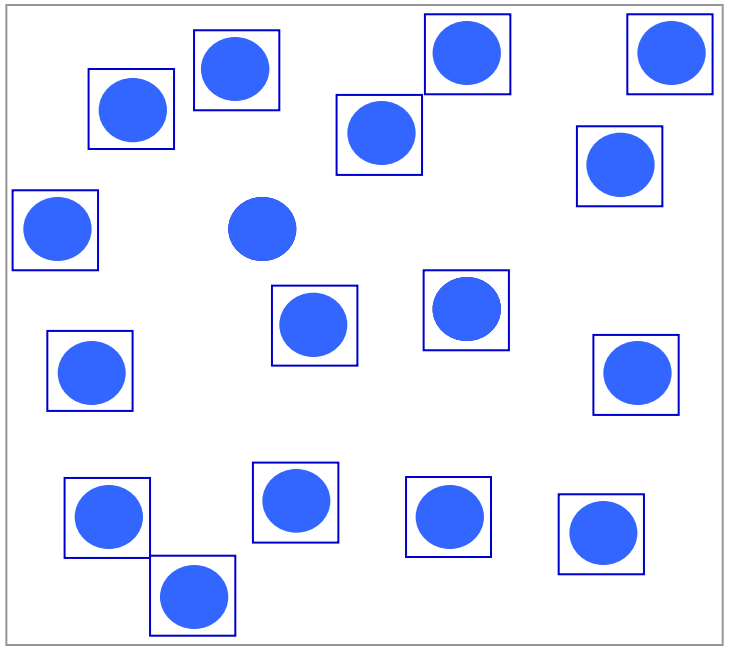


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Asymmetric preattentive factors

- Adding marks is better than removing them
  - it is better to underscore the word you want than underscoring everything but that word

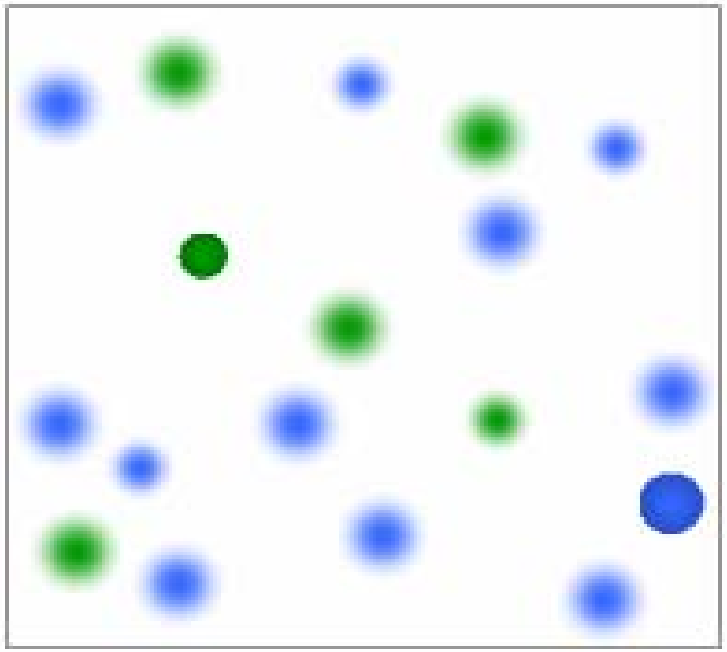


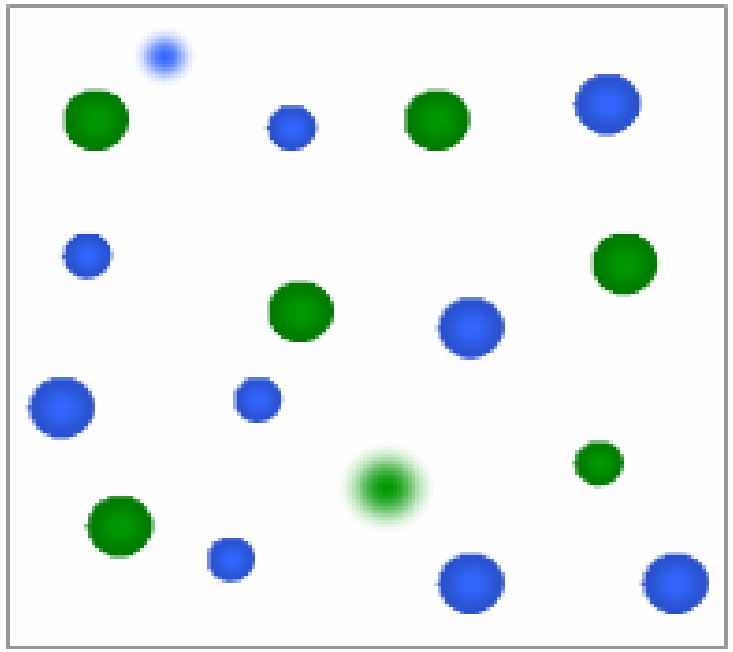


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Asymmetric preattentive factors

- Sharpness is asymmetrically preattentive
- Blurred shapes are much less highlighted





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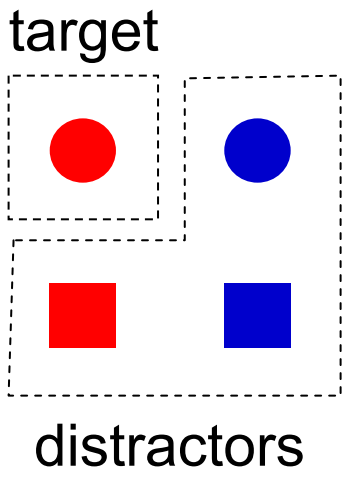
Degrees of popouts

- The strongest effects are based on color, orientation, size, contrast, and motion or blinking
  - this corresponds to the findings of neuropsychology
- Effects such as line curvature tend to be weaker
- Popouts are stronger when the distractors have uniform features
- Large color differences have more popout than small ones

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Conjunctions of features

- A conjunction target is a target made up of the intersection of non-unique features
- Examples
  - Position + Color
  - Size + Color
  - Position + Shape
  - Stereo + Color
  - Color + Motion

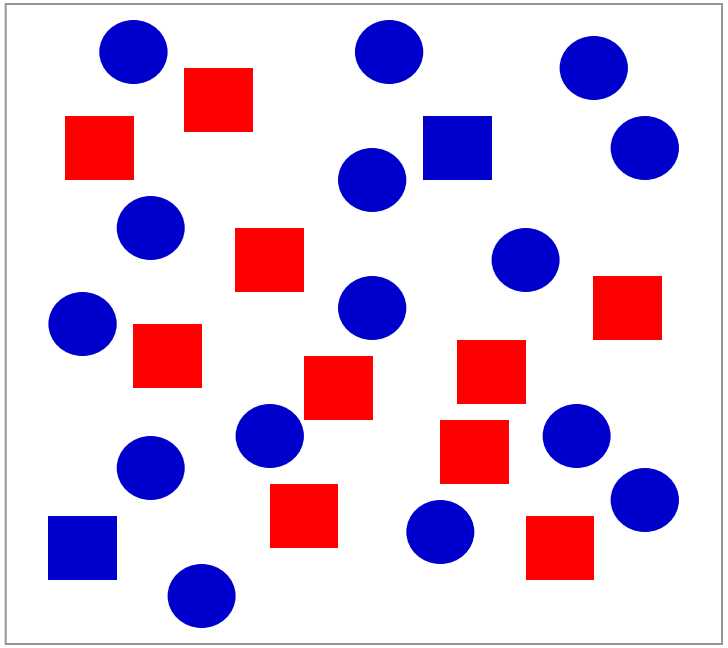


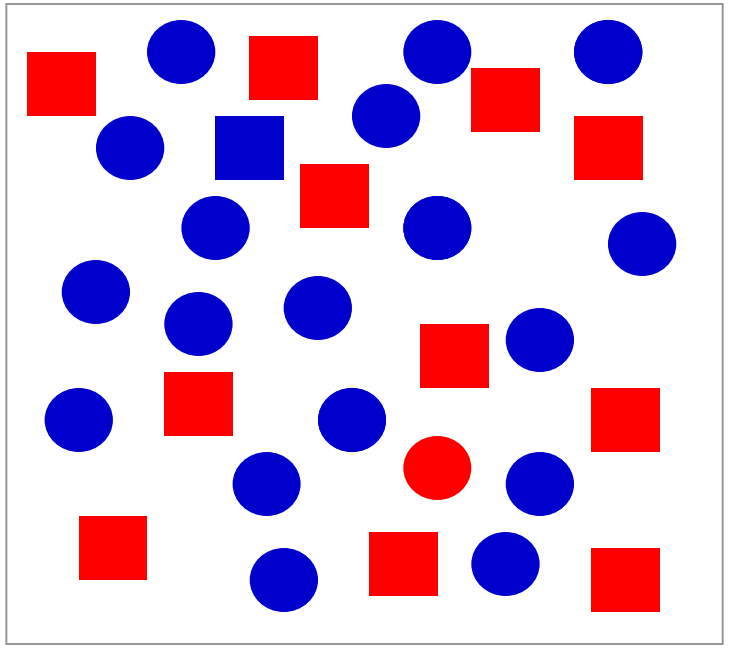
- Generally can not be preattentively detected

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Conjunction of hue and shape

- Search for the red circle ● among the many distractors ● ■ ■
  - conjunction of hue and shape does not pop out

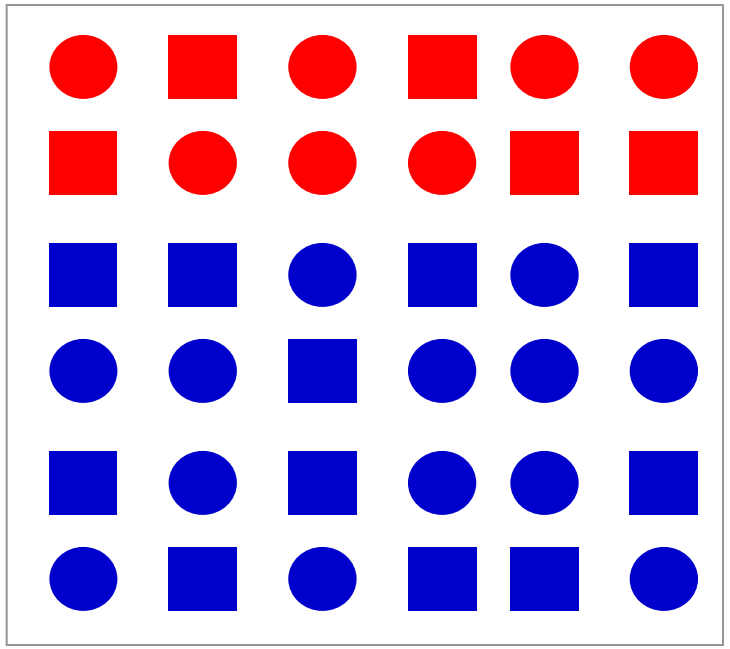


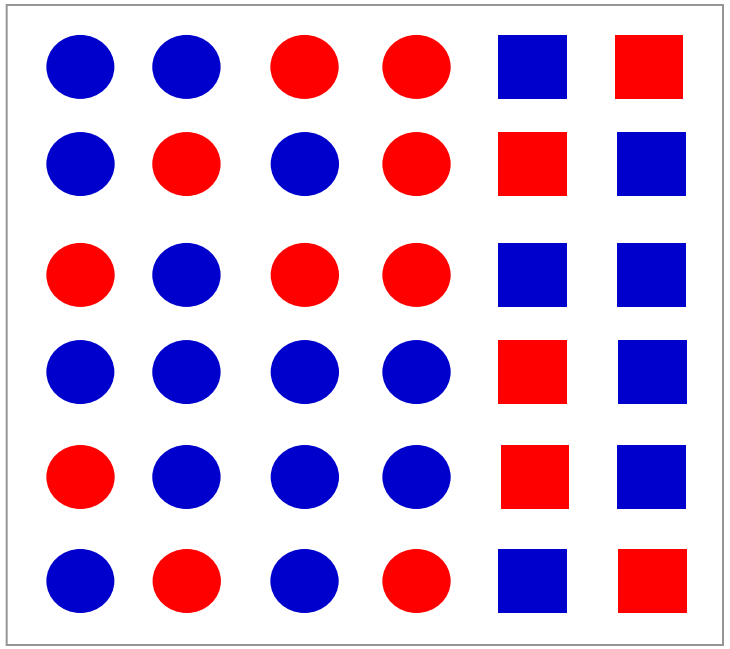


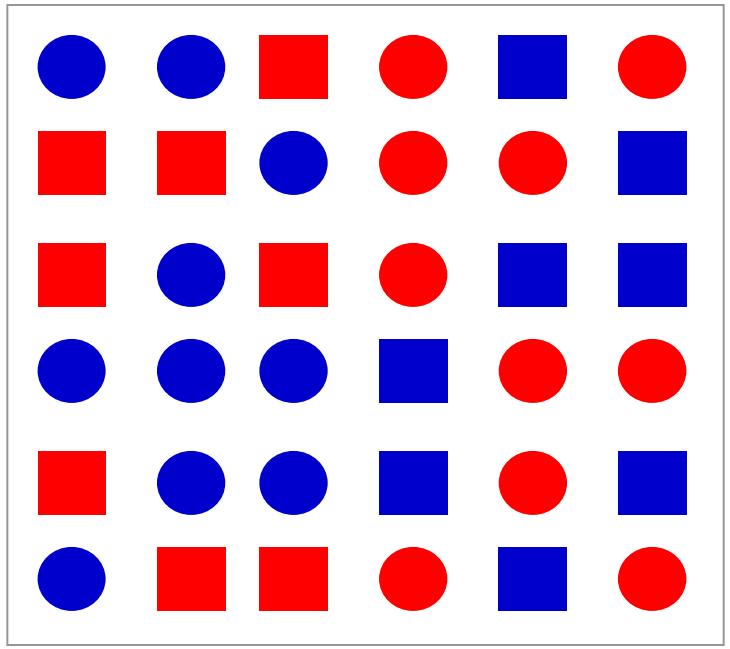
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Conjunction of hue and shape

- Find the boundary in this pictures
  - conjunction of hue and shape does not allow us to find the boundary



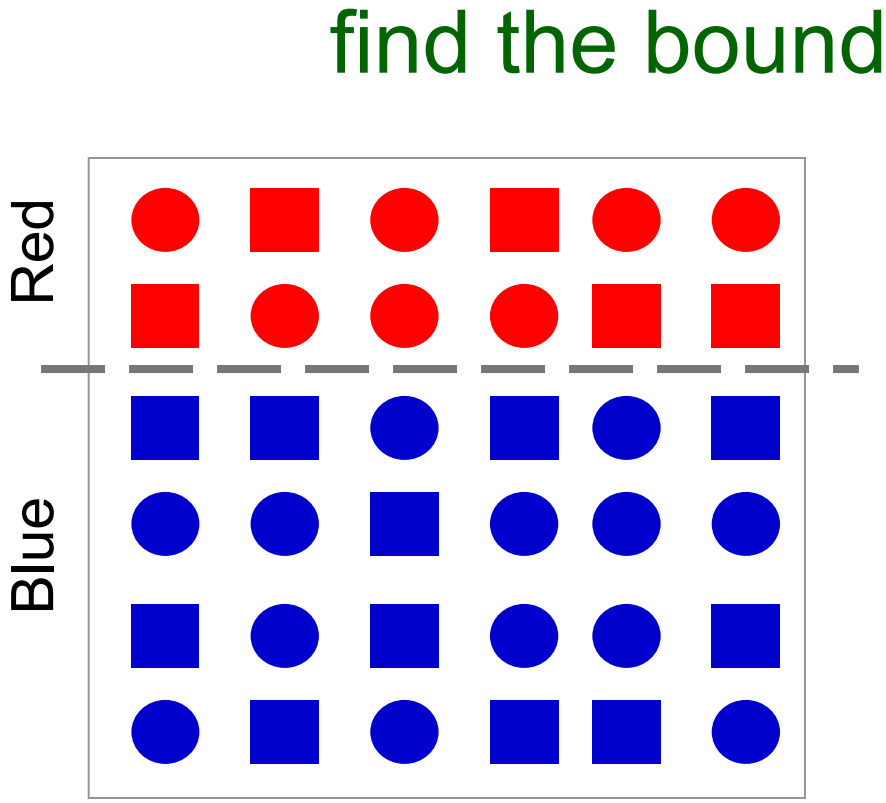


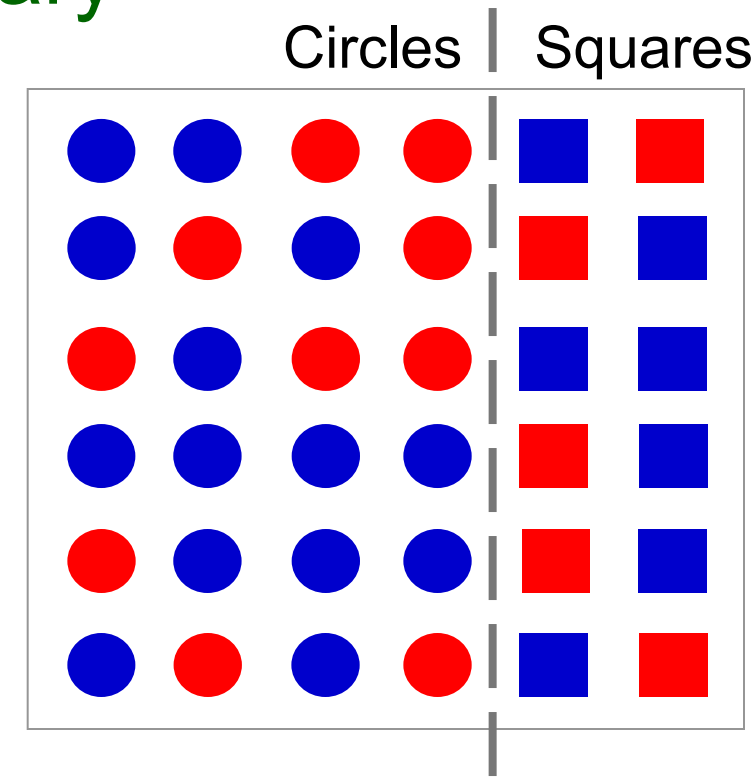


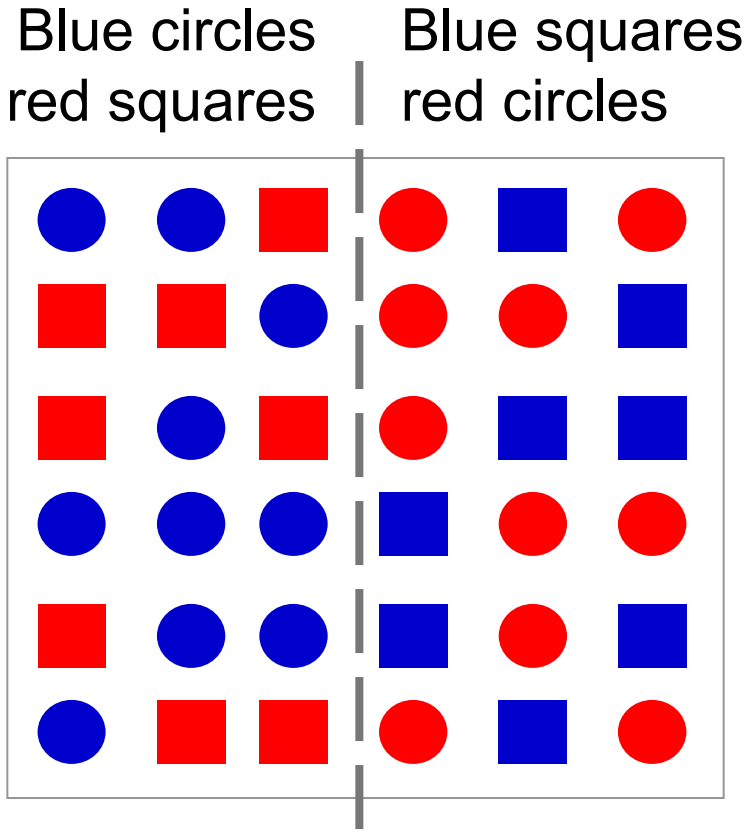
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Conjunction of hue and shape

- Find the boundary in this pictures
  - conjunction of hue and shape does not allow us to find the boundary







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General guidelines

- For maximum popout, a symbol should be the only object in a display that is distinctive on a particular feature channel
- To highlight some element
  - use positively asymmetric preattentive cues
  - use whatever feature is used least in other parts of the design
  - use motion sparingly
  - use strong preattentive cues before weak ones where ease of search is critical
  - avoid conjunction of attributes

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Interferences in perception

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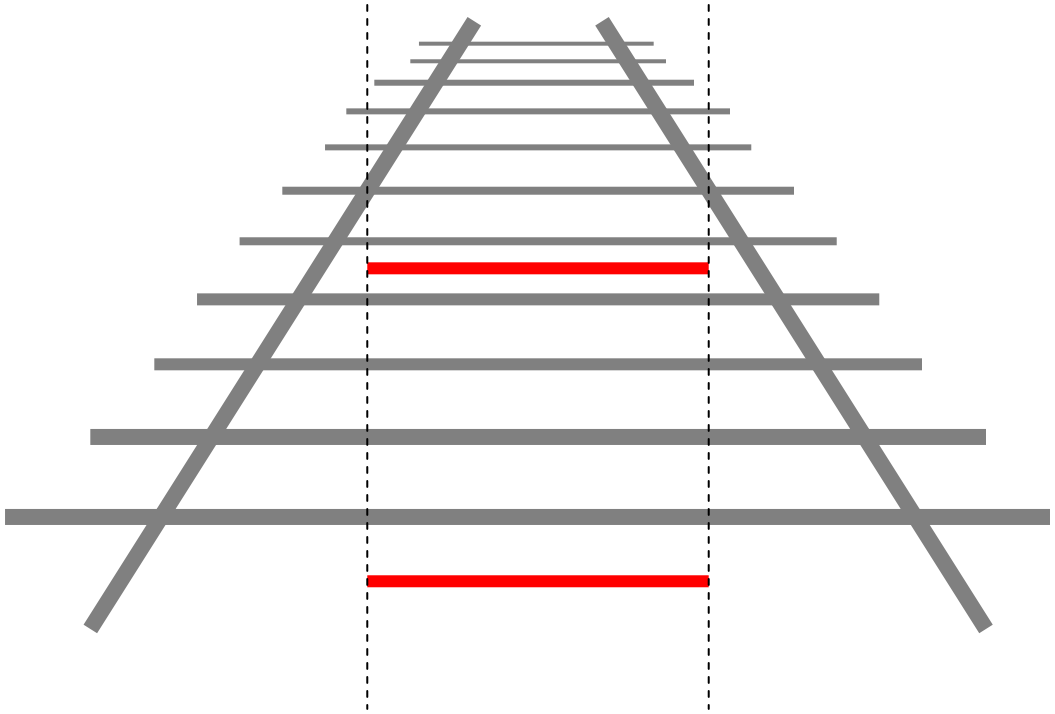
Interferences in perception

- In the following we will see that perceptive stimula may interfere with one another
  - lengths interfere with areas and vice versa
  - areas intefere with volumes and vice versa
  - lengths and areas interfere with their background
- perception is a comparative process

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The impact of perspective view

- Ponzo illusion
  - proposed by Mario Ponzo in 1911
  - shows that we judge an object's size based on its background



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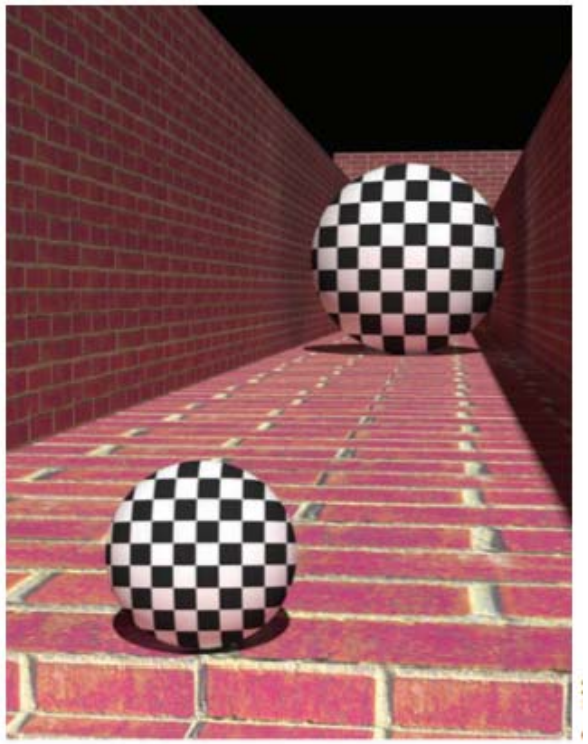
More on Ponzo illusion



Picture from wikipedia: “Moon illusion”

- Moon illusion
  - appears bigger when near the horizon

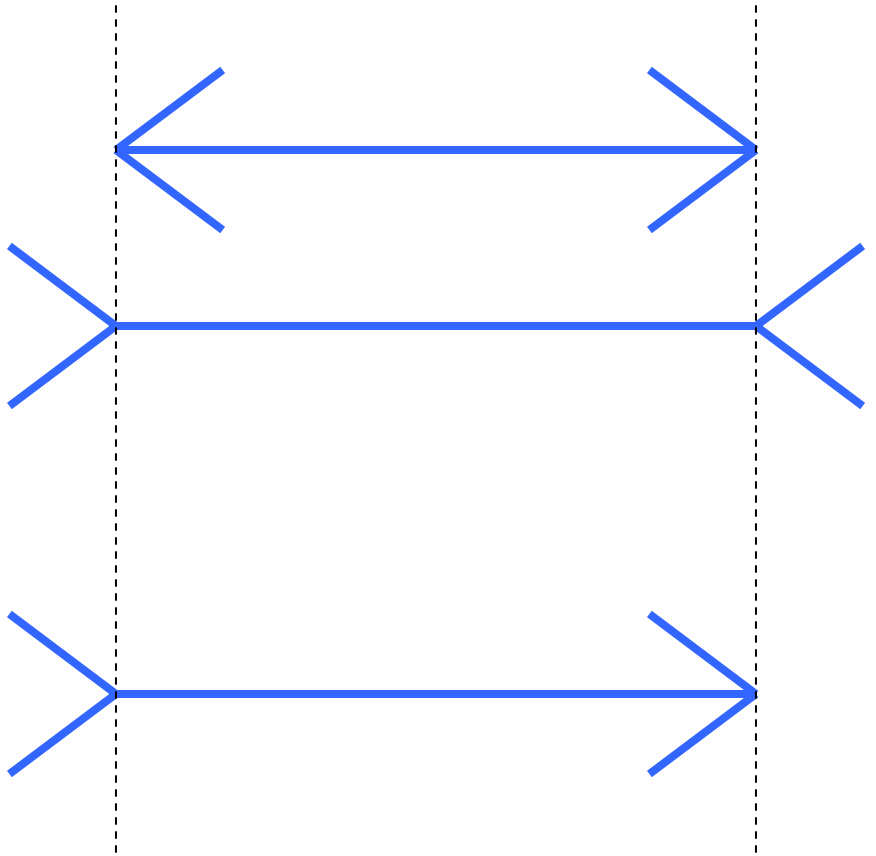
- Through functional magnetic resonance imaging (fMRI) [Murray et al 06] show that the retinotopic representation of an object changes in accordance with its perceived angular size



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Perception of lengths

- Müller-Lyer illusion
  - discovered by Franz Carl Müller-Lyer in 1889
  - shaft with two arrow tails seems longer than shaft with two arrow heads
  - may cause errors of 25-30%



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The perspective explanation

- Slanted lines are associated with perspective images
  - hypothesized by Richard Gregory in 1966

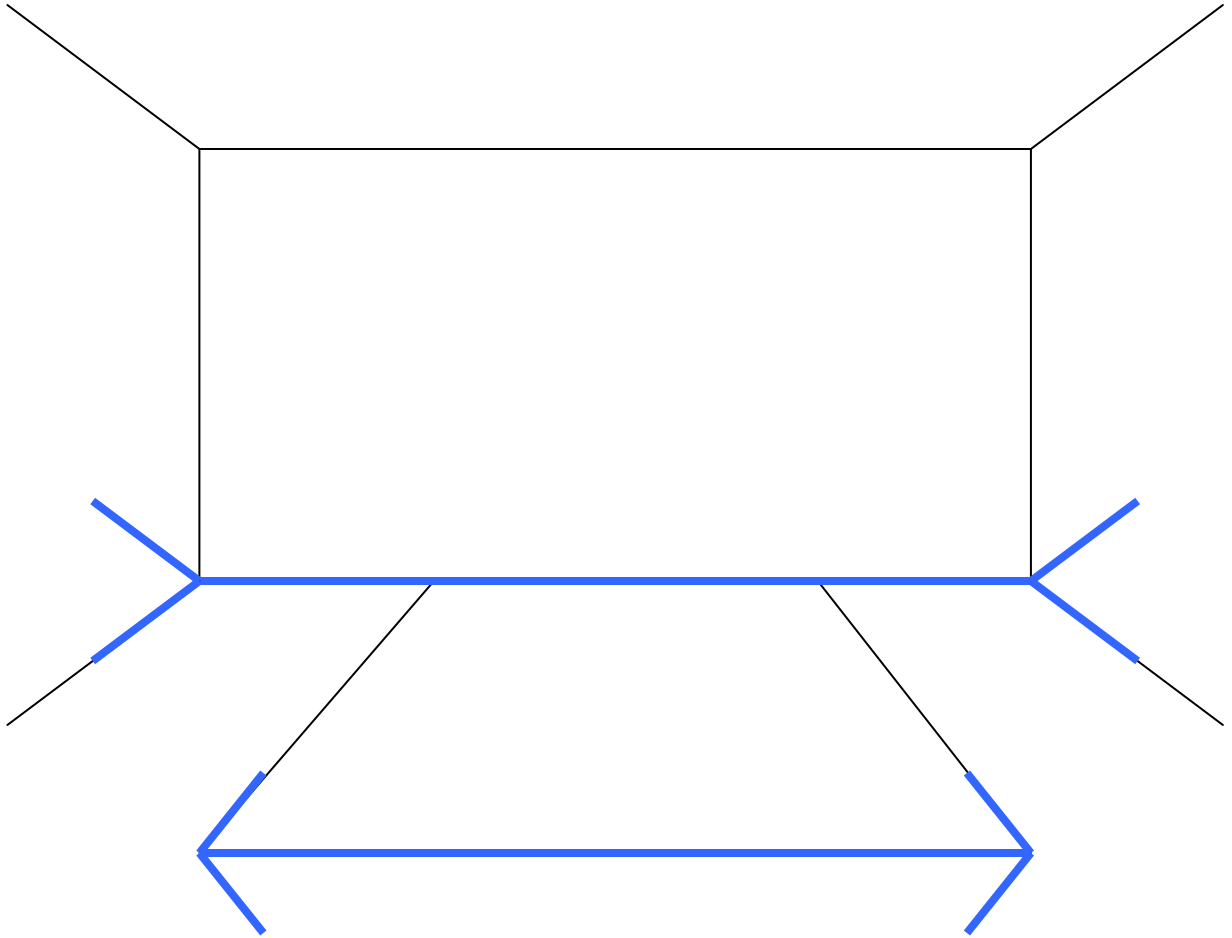
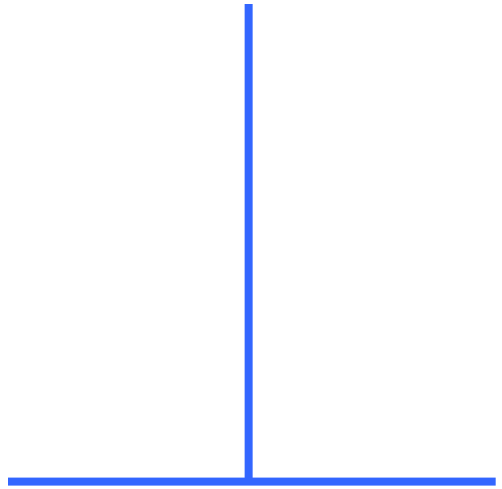


Image adapted from wikipedia: “Müller-Lyer illusion”

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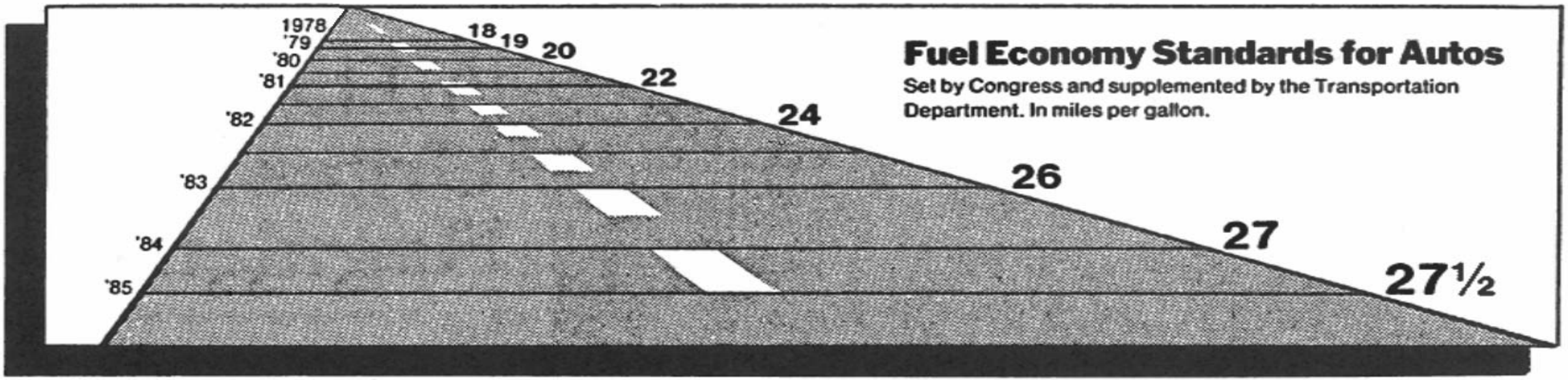
Vertical and horizontal perspective

- Horizontal/vertical illusion
  - discovered by Wundt in 1858
    - may cause a 30% error
  - we overestimate vertical lines with respect to horizontal ones
  - we overestimate bisecting lines with respect to bisected ones
  - also true for crosses and L-shaped configurations
    - rotated versions have minor effects



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Misleading use of perspective

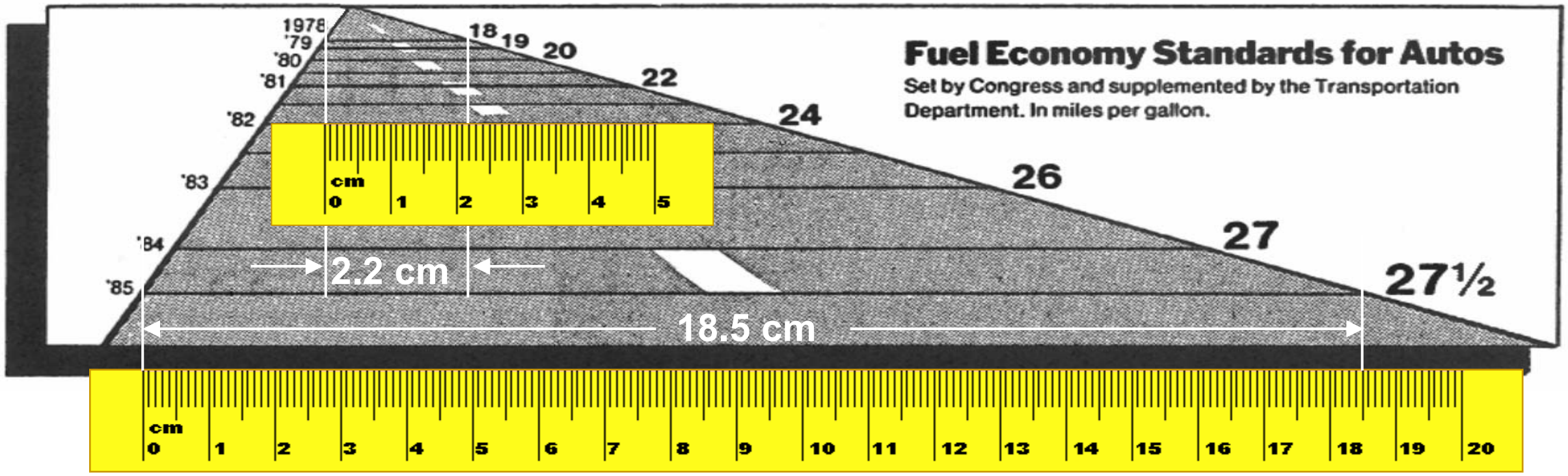


New Your Times, August 9, 1978, p.D-2

- Taking perspective into account
  - all horizontal lines should have the same length
    - From [Turfe 83]

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Misleading use of perspective

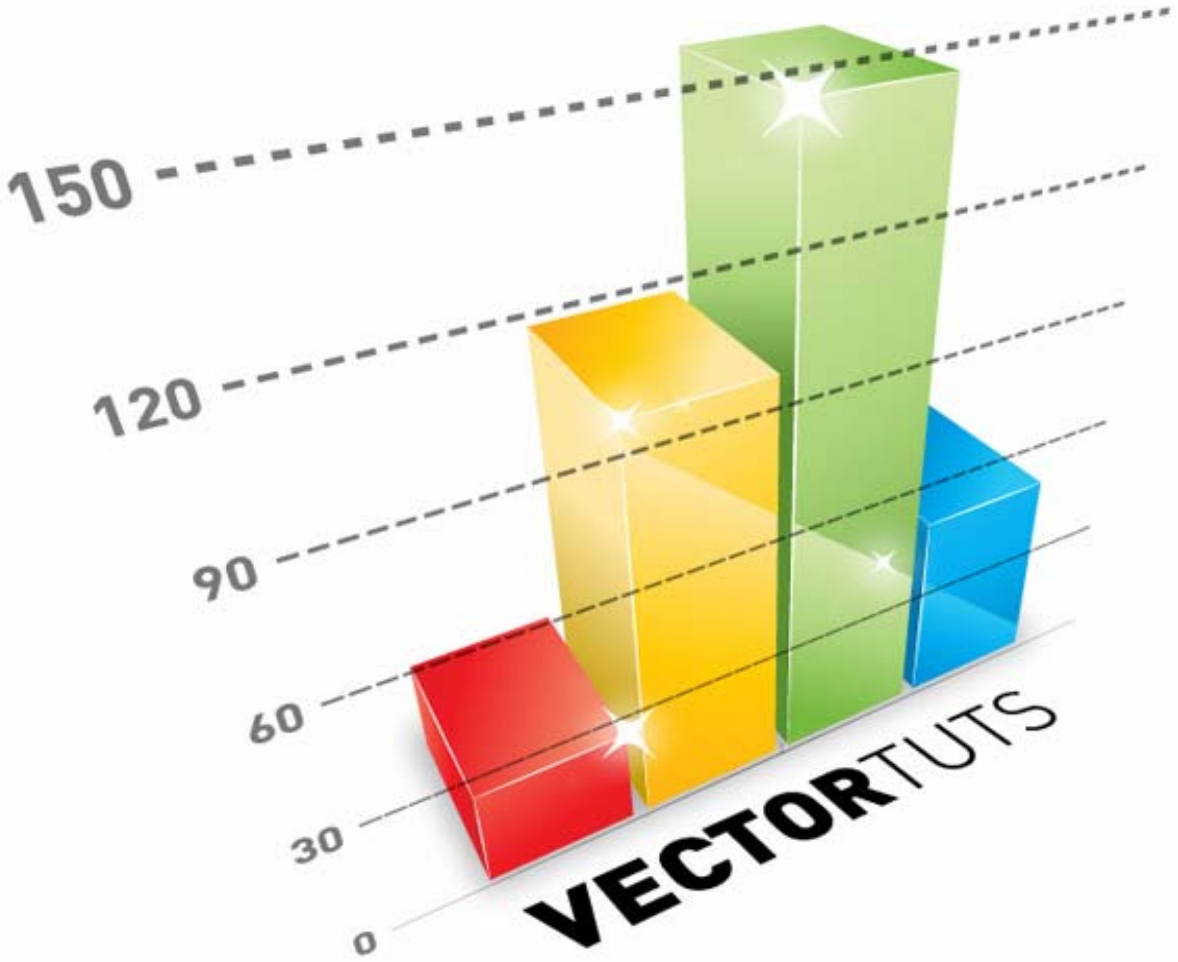


- Ignoring perspective
  - the line for 1985 is 18.5/2.2 = 8.4 times the line for 1978
  - the increase in the data, instead, is 27.5/18 = 1.53

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Misleading use of perspective

- What is the size of the red bar?
  - 25?
  - 29?

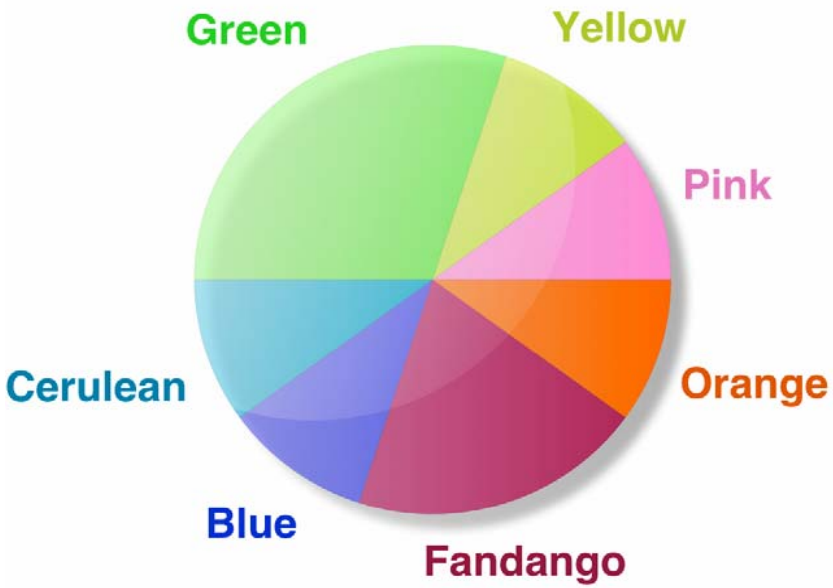


http://design.tutsplus.com/tutorials/turn-a-boring-bar-graph-into-a-3d-masterpiece--vector-537

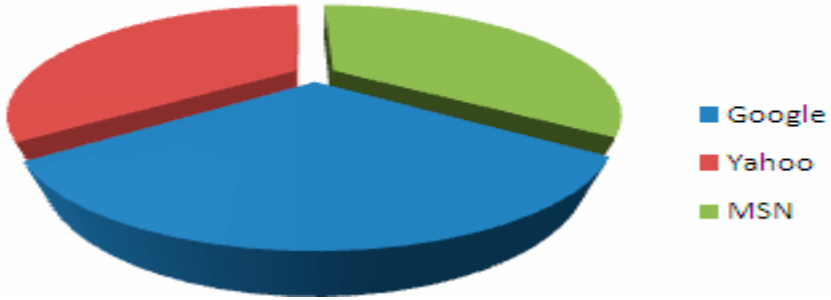
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Misleading use of perspective

- A good example
  - 3D flavour is used for improving aesthetics
- A bad example
  - 3D perspective deceives the user perception
    - Google = 30%
    - Yahoo = 35%
    - MSN = 35%



http://bynomial.com/blog/?p=104



http://infolific.com/technology/internet/seo-lie-factor/

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Misleading use of perspective

- A or B, which bar is taller?

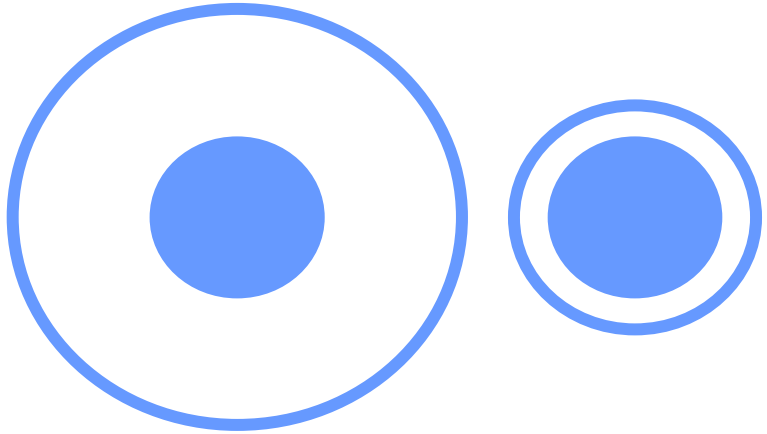


http://freedesignfile.com/35792-business-3d-graph-vector-material-05/

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Relative size perception

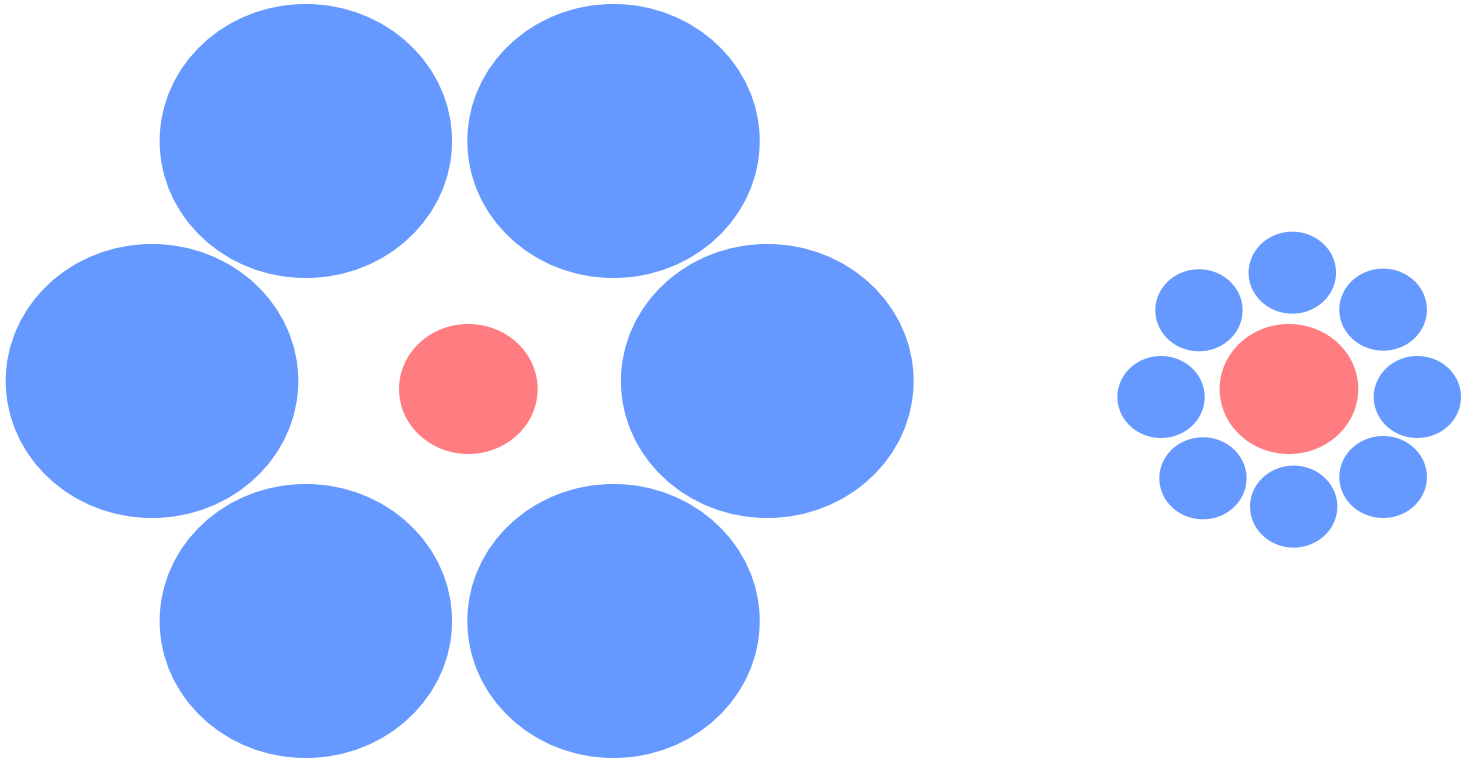
- Delboeuf illusion
  - discovered by Joseph Remi Leopold Delboeuf in 1887-88
  - the surrounded circle appears larger than the non-surrounded one if the annulus is close, smaller if the annulus is distant



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Relative size perception

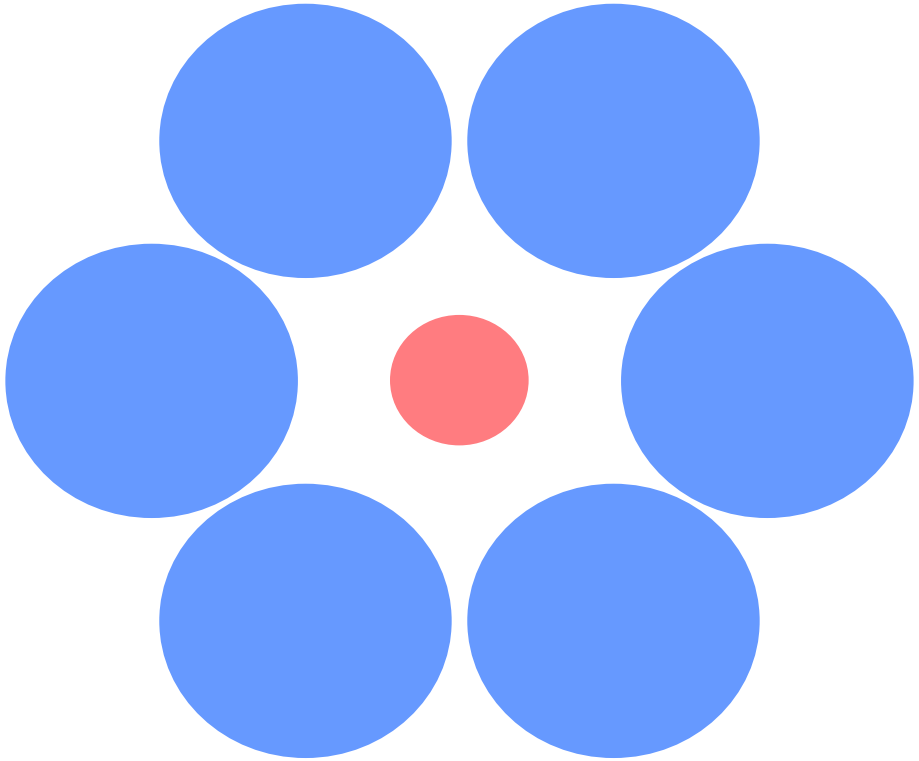
- Ebbinghaus illusion
  - discovered by Hermann Ebbinghaus (1850-1909)
    - popularized by Edward B. Titchener in 1901



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Dynamic Ebbinghaus

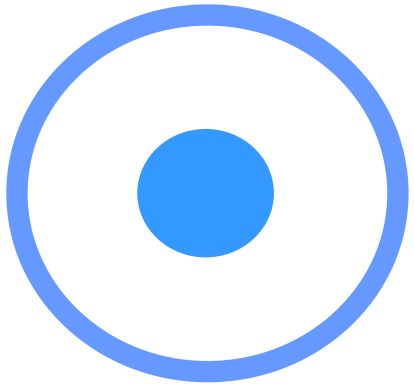
- The illusion is enhanced in the dynamic version
  - [Mruczek et al 15]



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Dynamic Delboeuf

- Is the illusion is enhanced in the dynamic version?



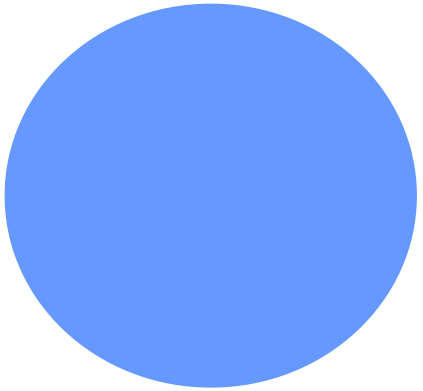
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Measuring areas is difficult

- How much bigger is the second circle?
  - ☐ 1.2 times the area of the smaller one
  - ☐ 1.4 times the area of the smaller one
  - ☐ 1.6 times the area of the smaller one
  - ☐ 1.8 times the area of the smaller one
  - ☑ 2 times the area of the smaller one



Area = 300

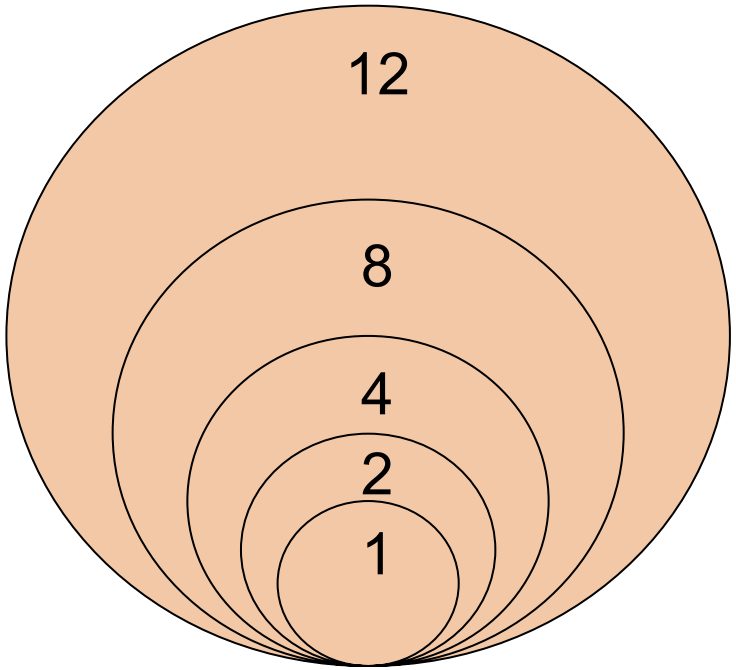


Area = 600

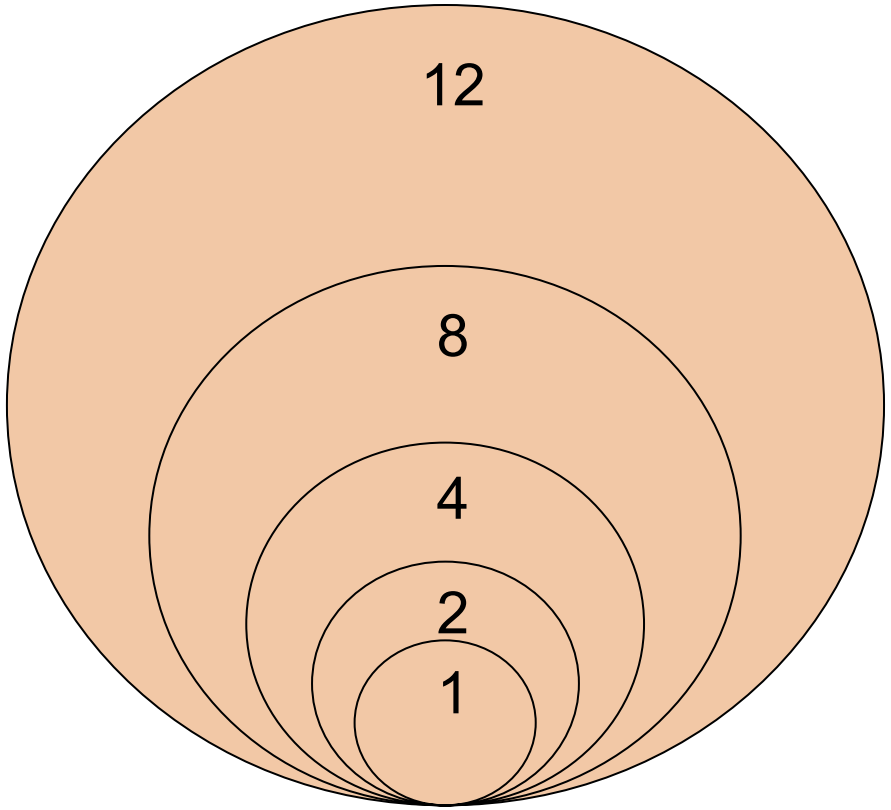
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Relative area perception

- Accounting for our bias



Circles drawn by absolute scaling



Circles drawn by apparent scaling [Flannery 71]

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When using circular areas

- A value *v* is supposed to be proportional to the area of a circle of radious *r*

$$v = c\pi r^2$$

  - c is an arbitrary constant

- This implies that you use the radious

$$r = bv^{0.5}$$

  - where $b = 1/(c\pi)^{0.5}$
  - since *c* was an arbitrary constant, *b* can be arbitrarily choosen as well

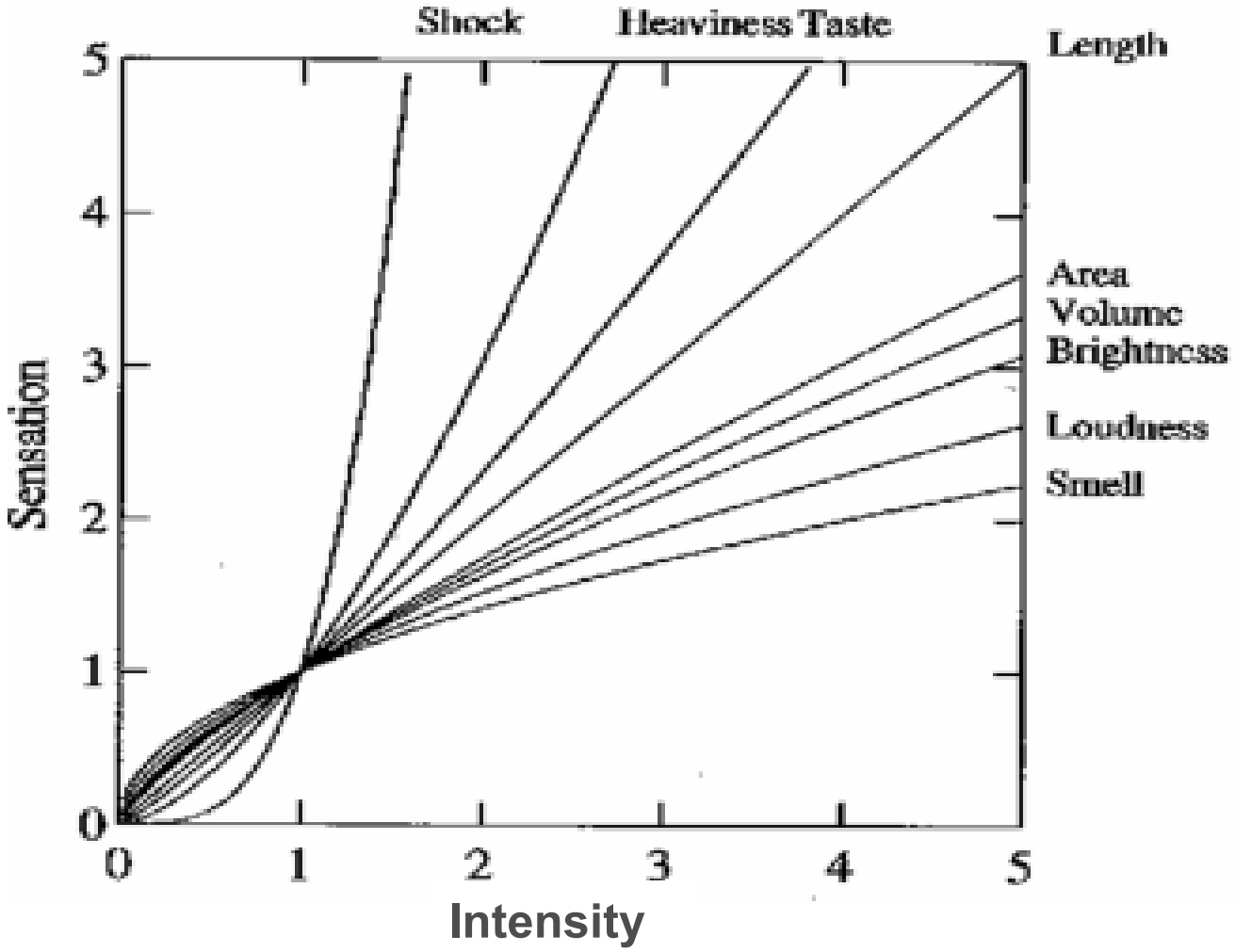
- Flannery, after experimentation, concluded that is preferable to use the radious

$$r = bv^{0.57}$$

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Stevens' power law

- $s(x) = cx^b$
- where
  - s = sensation
  - x = intensity of the stimulus
  - c = multiplicative constant
- overestimate
  - b > 1
- underestimate
  - b < 1



[graph from Wilkinson 99, based on Stevens 61]

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Stevens' power law

- Exponent *b* for several sensations

Sensation	Exponent
Brightness	0.33 (5° target) – 0.5 (point source)
Loudness	0.54 (monoaural) – 0.6 (binaural)
Smell	0.55 (Coffee) – 0.6 (Heptane)
Taste	0.8 (Saccharine) – 1.3 (Salt)
Temperature	1.0 (Cold) – 1.6 (Warm)
Vibration	0.6 (250 Hz) – 0.95 (60 Hz)
Duration	1.1
Heaviness	1.45
Electic Shock	3.5

[Stevens 61]

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Stevens' power law

- Experimental results for *b*

Sensation	Exponent
Length	0.9 – 1.1
Area	0.6 – 0.9
Volume	0.5 – 0.8

- Heuristic: b ~ 1/sqrt(dimensionality)
  - lengths ⇒ b ~ $1/\sqrt{1} = 1$
  - areas ⇒ b ~ $1/\sqrt{2} = 0.7$
  - volumes ⇒ b ~ $1/\sqrt{3} = 0.58$

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Steven's power law in practice

- Let's focus on areas
- We want a perception s proportional to a value v : $s \approx v$
- Since $s \approx x^b = x^{\frac{1}{\sqrt{2}}}$
- We need an area $x \approx s^{\sqrt{2}} = s^{1.414} \approx v^{1.414}$
- Hence we could use a radious

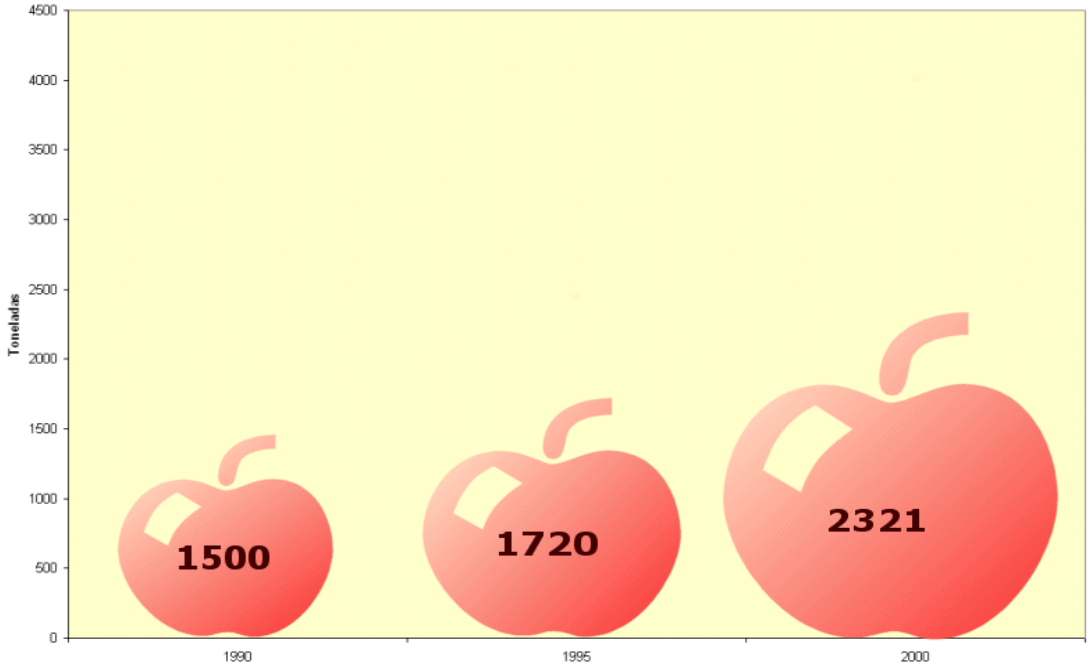
$$r \approx \left(v^{1.414}\right)^{0.5} = v^{0.707}$$

  - slightly higher than the value racommanded by Flannery

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Misleading use of area

- The scale encodes lengths but the representation shows areas
  - the height of each apple is proportional to the value
  - between the first and last value there is a difference of 54% but the difference in area is 139%
    - from http://www.infovis.net/printMag.php?lang=2&num=109



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Turfe's graphical integrity

- Graphic (as text) may lie
  - unintentional lapses
  - intentional rhetorical exaggeration
- Given a graphic and the data it represents we can assign to it a *lie factor*

$$\text{lie factor} = \frac{\text{size of the effect in graph}}{\text{size of the effect in data}}$$

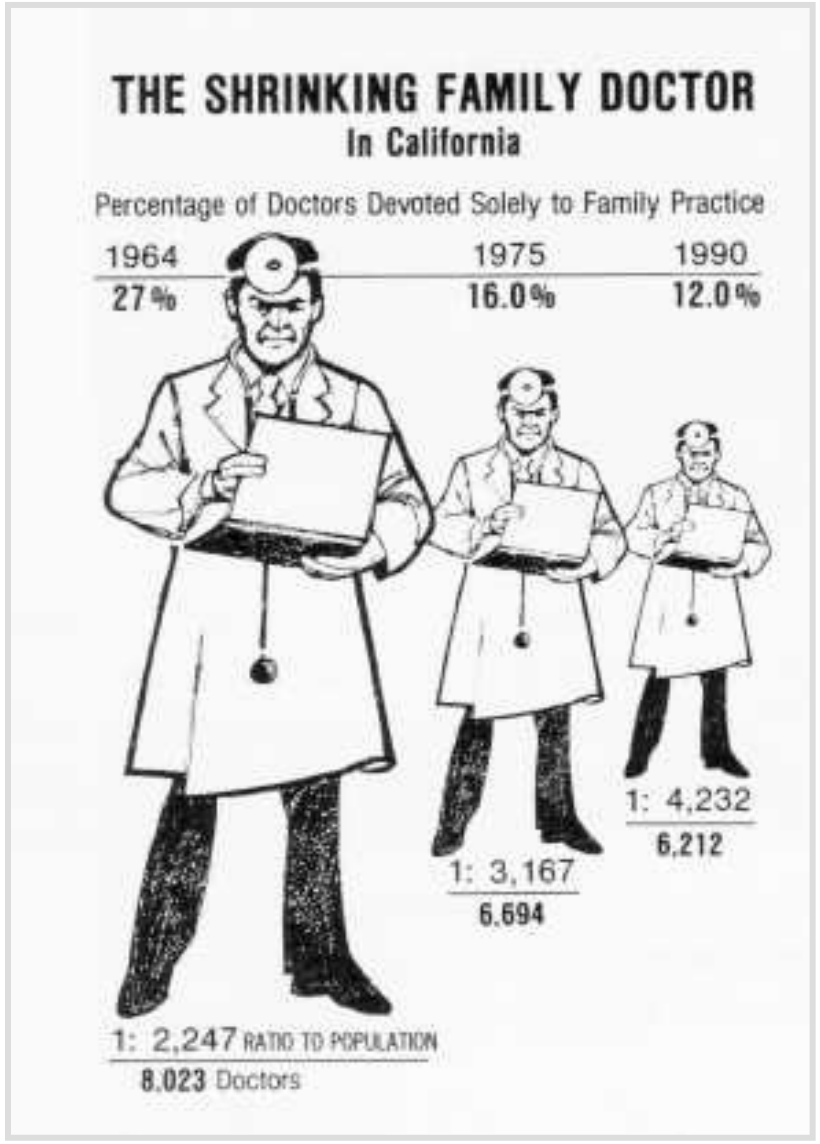
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Turfe's graphical integrity

- Main causes of lies in graphs
  - misleading uses of area or volume
  - misleading uses of perspective
  - leaving out important context
  - lack of taste and aesthetics
  - …
- Correct representation has lie factor = 1
  - "a lie factor greater than 1.05 or less than 0.95 indicates a substantial distortion, far beyond minor inaccuracies“

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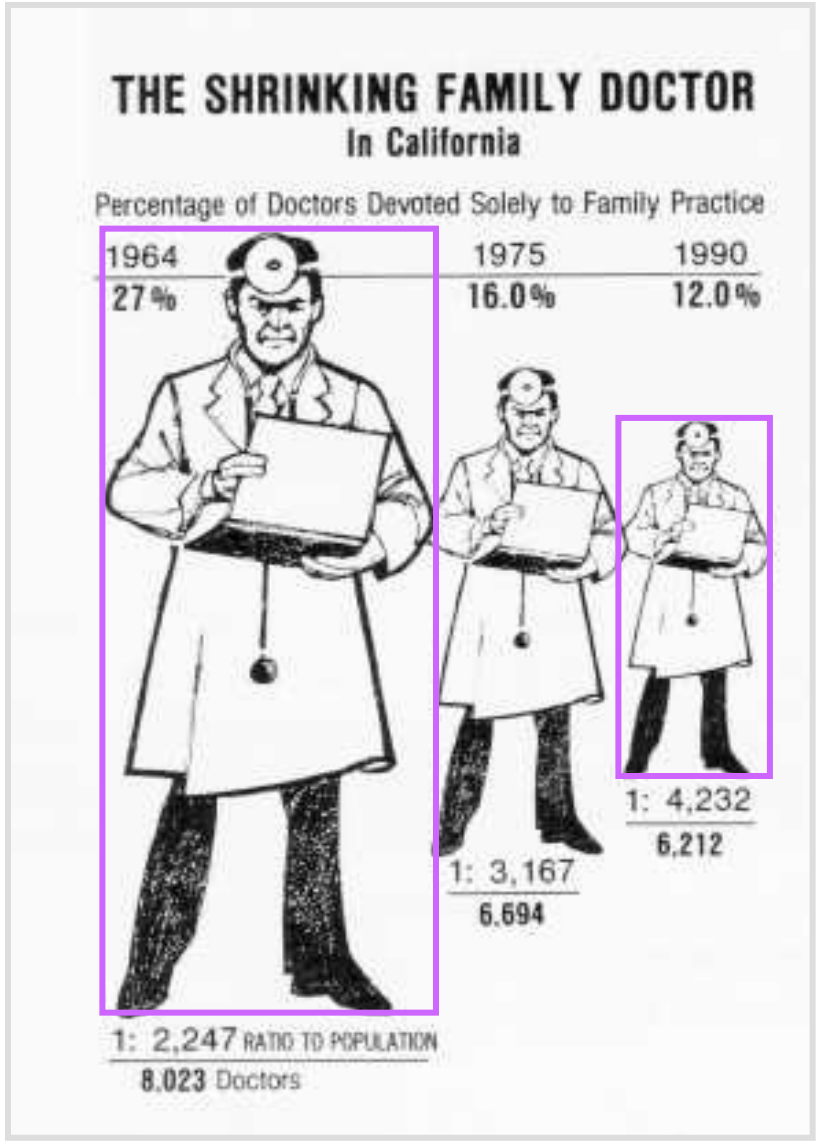
Misleading use of area



Los Angeles Times, August 5, 1979, p.3

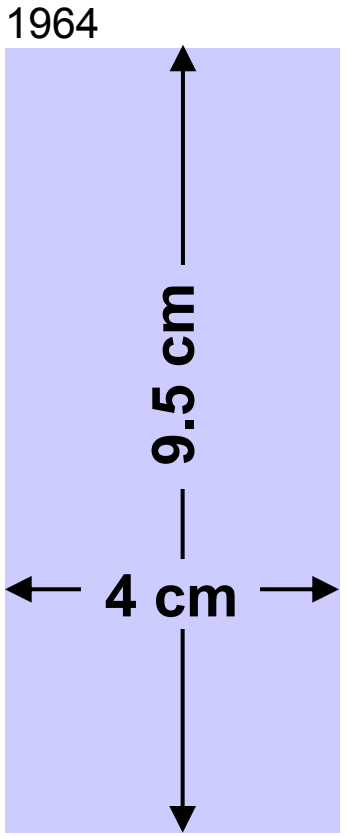
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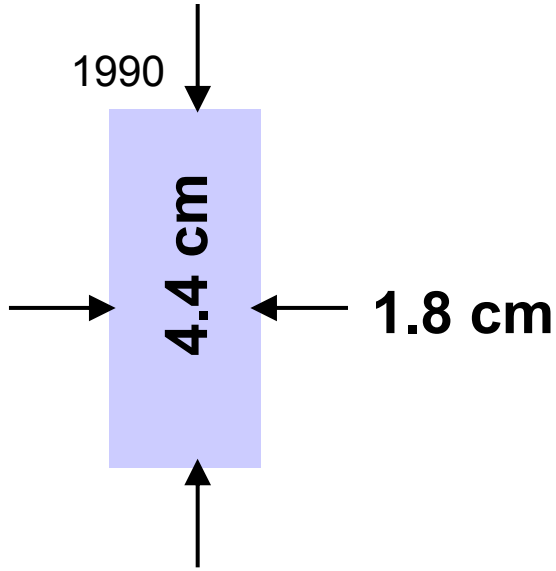
Misleading use of area



Los Angeles Times, August 5, 1979, p.3

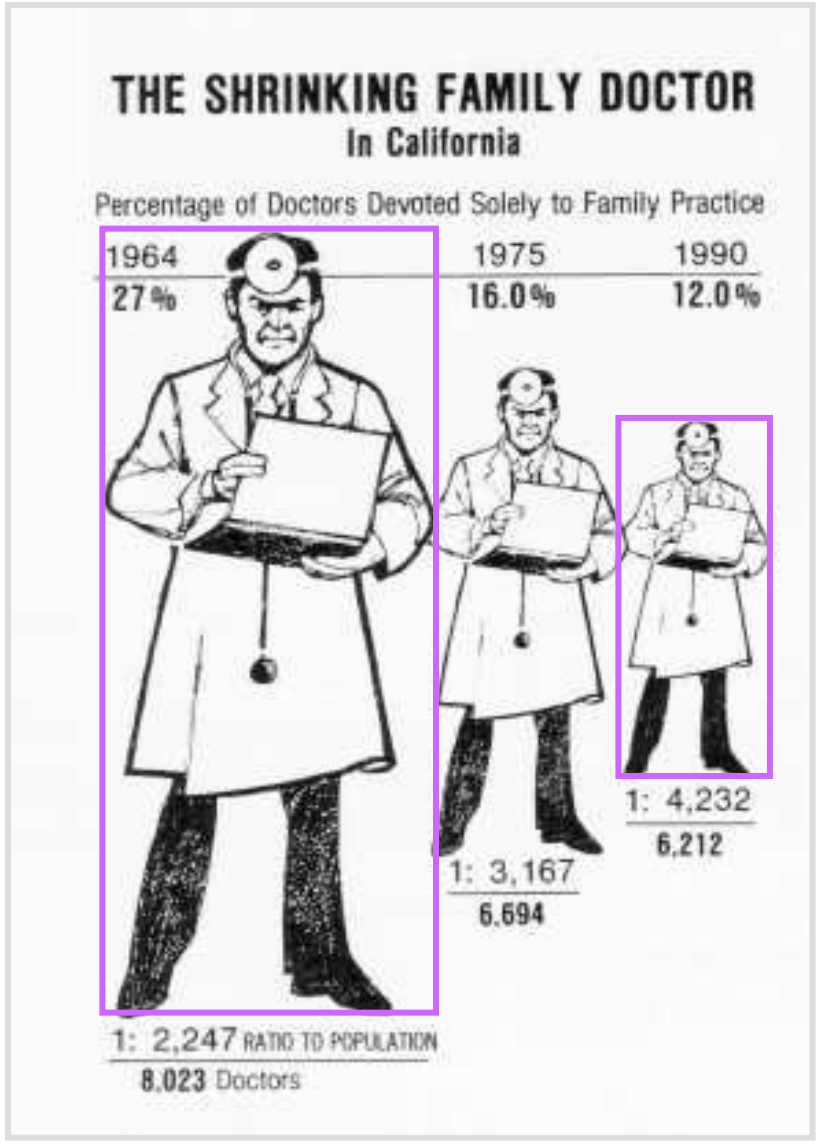
- 1964 area = 38 cm^2
- 1990 area = 7.92 cm^2





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Misleading use of area



Los Angeles Times, August 5, 1979, p.3

- Size of the effect in the graphics

  (38 – 7.92)/7.92 = 3.79 times bigger

- Size in data

  (27 – 12)/12 = 1.25 times bigger

- Lie factor

  3.79/1.25 = 3.032

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Misleading use of area



- Varying both dimensions
  - the bottom 1978 dollar should be twice as big
    - From [Tufte 83]

Washngton Post, October 25, 1978, p.1

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Misleading use of dimensions

- From [Turfe 83]
  - the use of two (or three) varying dimensions to show one-dimensional data is a weak and inefficient technique
  - the number of information-carrying dimensions should not exceed the number of dimensions in the data

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Weber's law

- In the 1830's, Weber made measurements of the just-noticeable differences (JNDs) in the perception of weight and other sensations
- He found that for a range of stimuli, the ratio of the JND ΔS to the initial stimulus S was relatively constant

JND ⟶ stimulus ⟶ $\frac{\Delta S}{S} = k$

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Weber's law

- Ratios are more important than magnitude in stimulus detection
  - for example, we detect the presence of a change from 100 cm to 101 cm with the same probability as we detect the presence of a change from 1 to 1.01 cm
    - even though the discrepancy is 1 cm in the first case and only .01 cm in the second

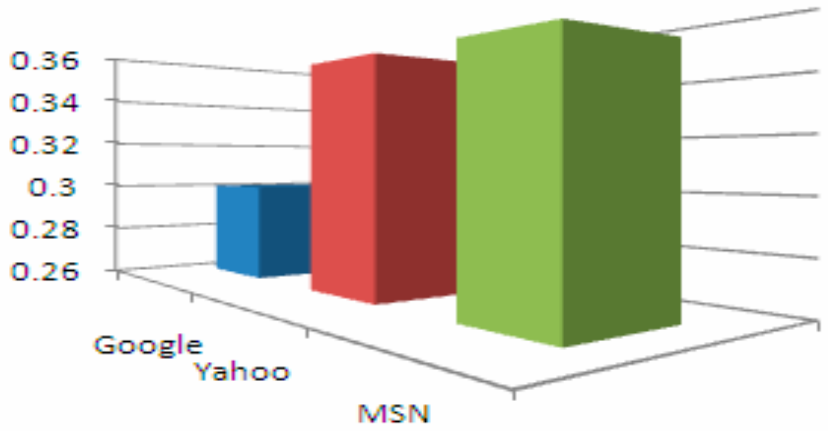


Blue and red bars discrepancy is the same, but with different ratios

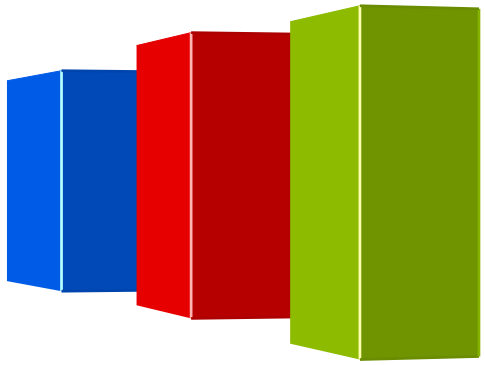
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Playing with scales

- Using non-zero scales (or non-equispaced scales) causes misinterpretations
  - Google = 30%
  - Yahoo = 35%
  - MSN = 35%



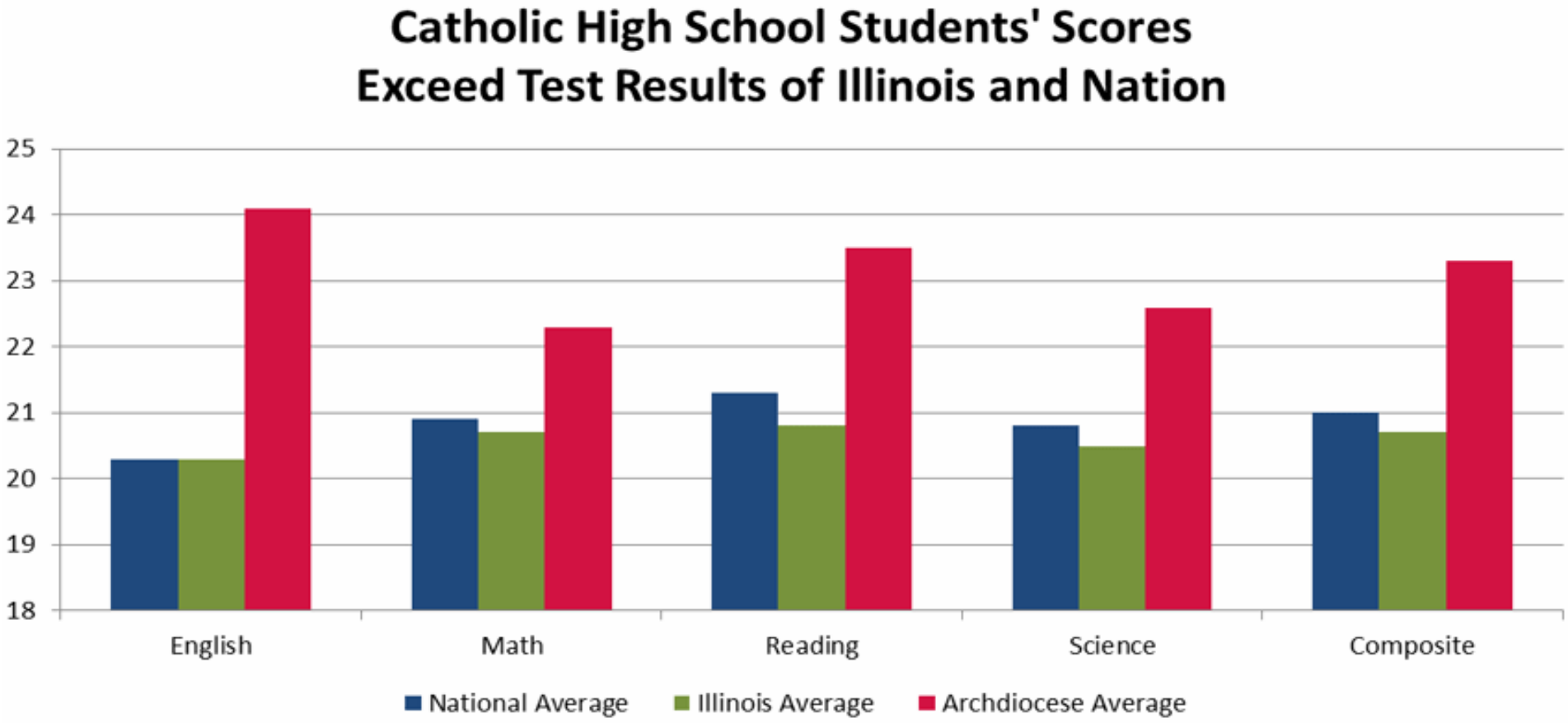
http://infolific.com/technology/internet/seo-lie-factor/



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A real example

- Are catholic students far better?



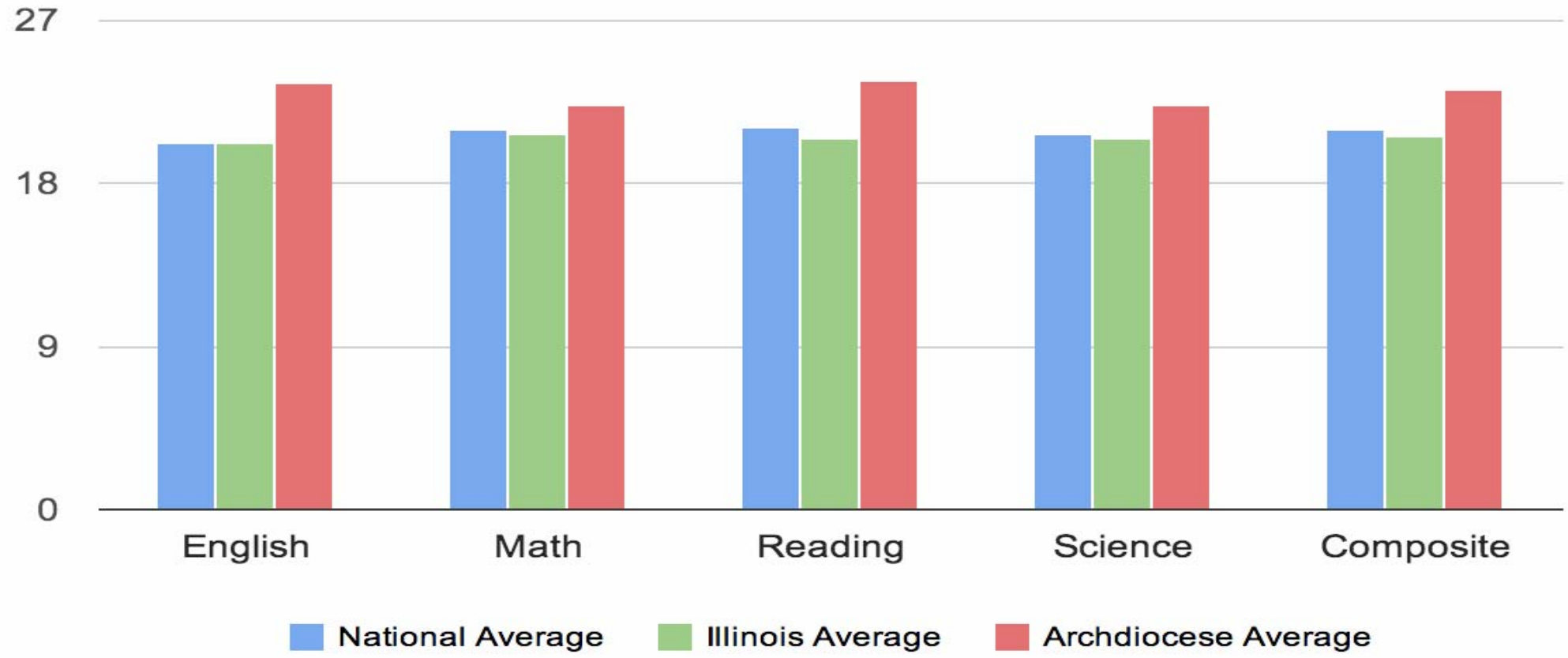
http://schools.archchicago.org/Academics/Charts/ACT.aspx

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A real example

- The same data using correct scales

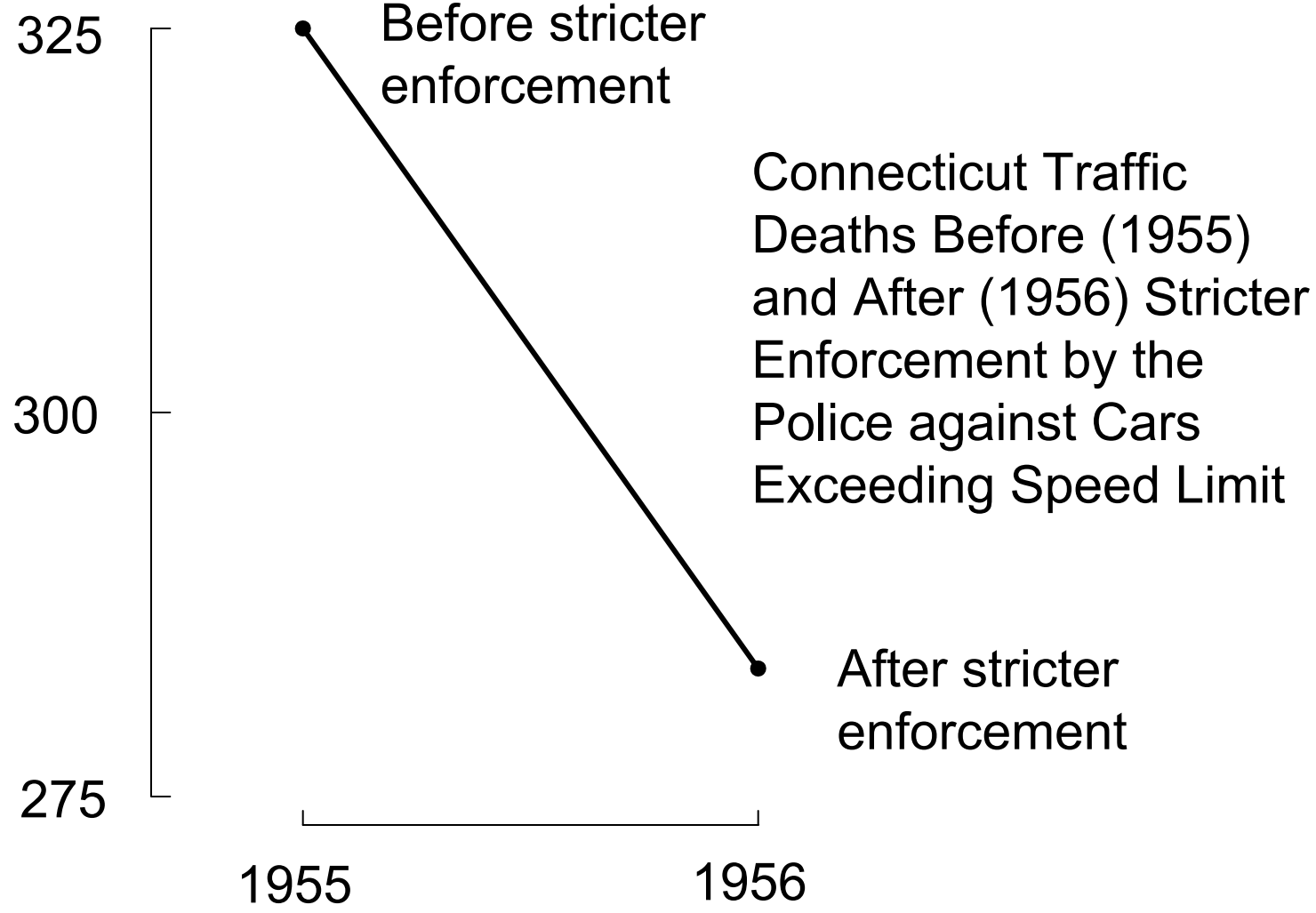
http://fitz.blogspot.it/2013/08/the-lie-factor.html



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Leaving out context

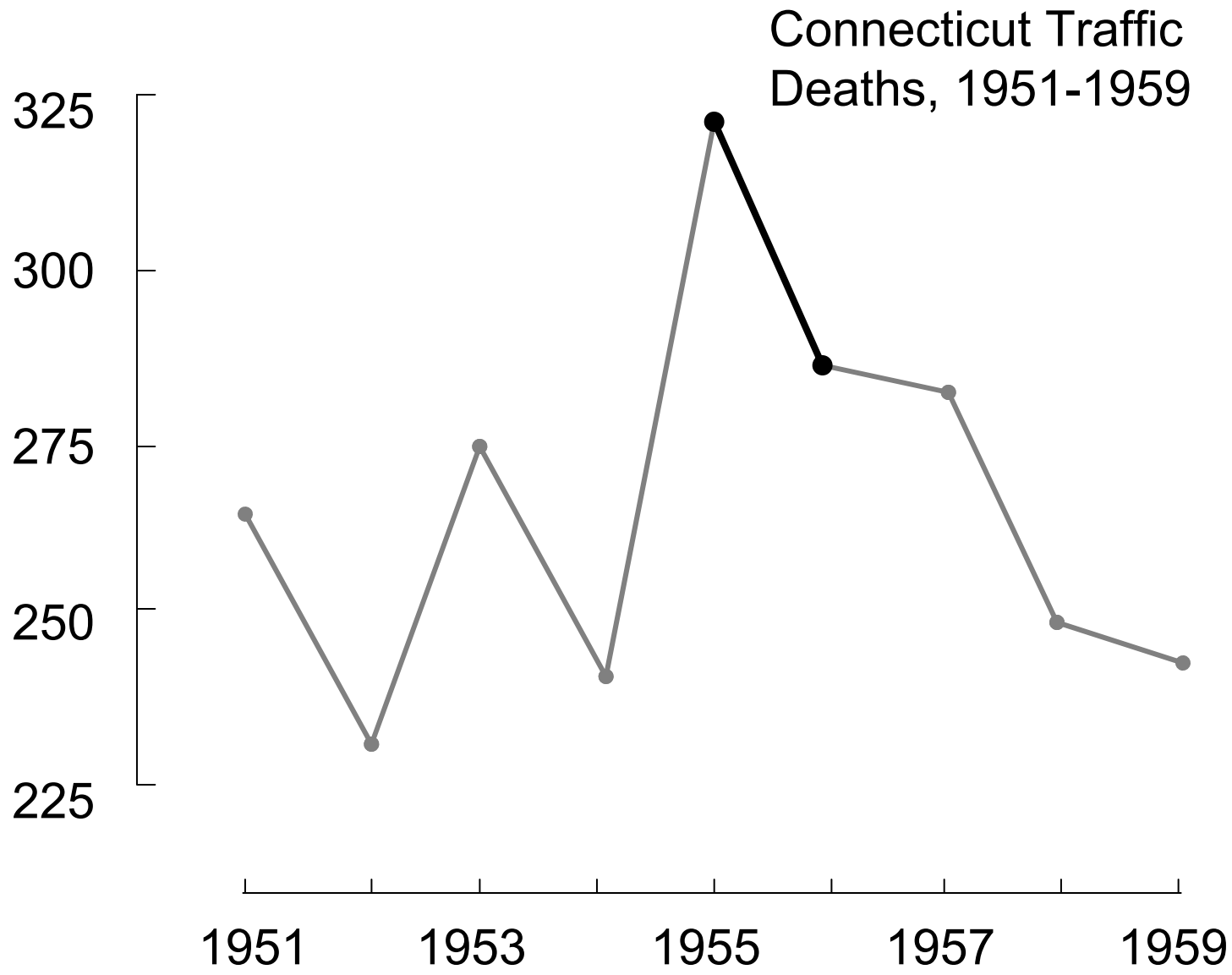
- Does the enforcement of speed limit reduce traffic deaths?
  - example from [Turfe 83]



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Adding some context

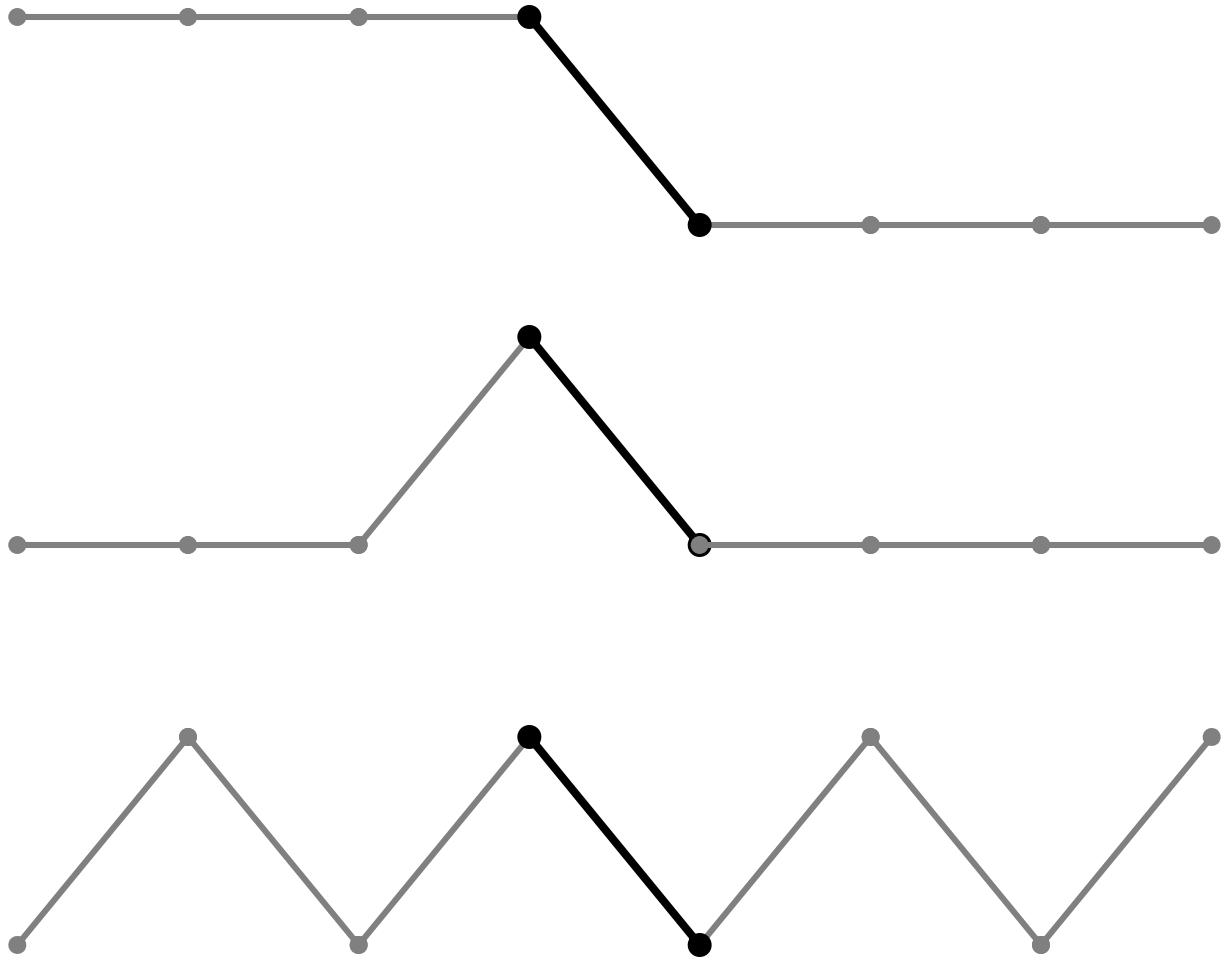
- 1955-1956 was not the only year when traffic deaths decreased



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Context helps interpreting data

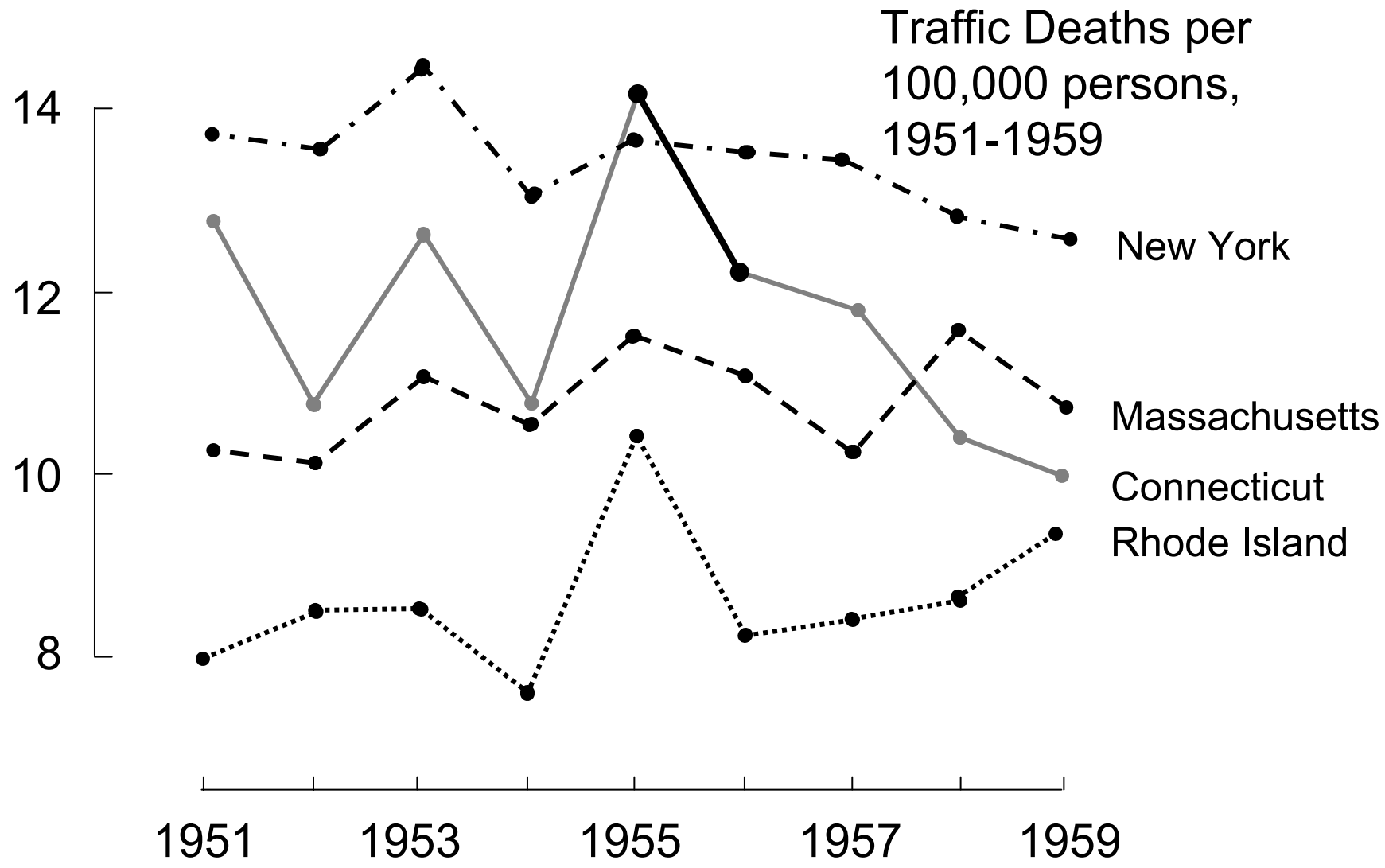
- Different patterns would give rise to different interpretations
  - definitive change?
  - transient phenomenon?
  - recurrent pattern?



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Adding more context

- In ’55-’56 there was a general reduction of traffic deaths
  - possibly due to a spending breakdown



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Mapping variables to graphics

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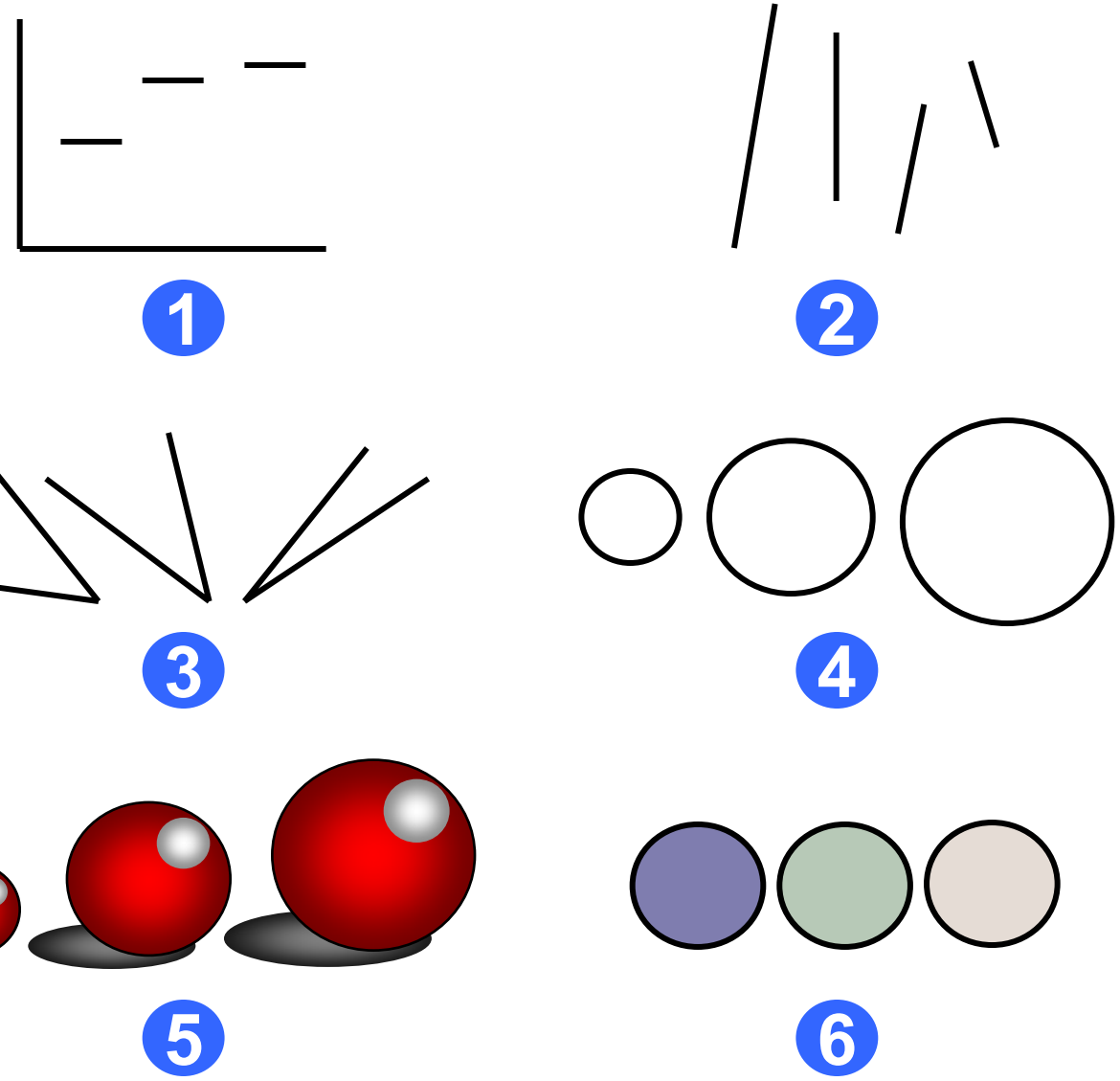
Mapping quantitative values

- Quantitative values are usually mapped to geometric objects
- Several alternatives
- A list sorted by perceptual accuracy [Cleveland & McGill 84]
    1. position
    2. length
    3. angle/slope
    4. area
    5. volume
    6. color/density

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Mapping quantitative values

1. Position
   - monodimensional
2. Length
3. Angle/Slope
4. Area
5. Volume
6. Color/Density



Adapted from [Hart 15]

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Mapping quantitative values

- Justification of the empirical results

1. Position
2. Length
3. Angle/Slope

one dimensional values

4. Area

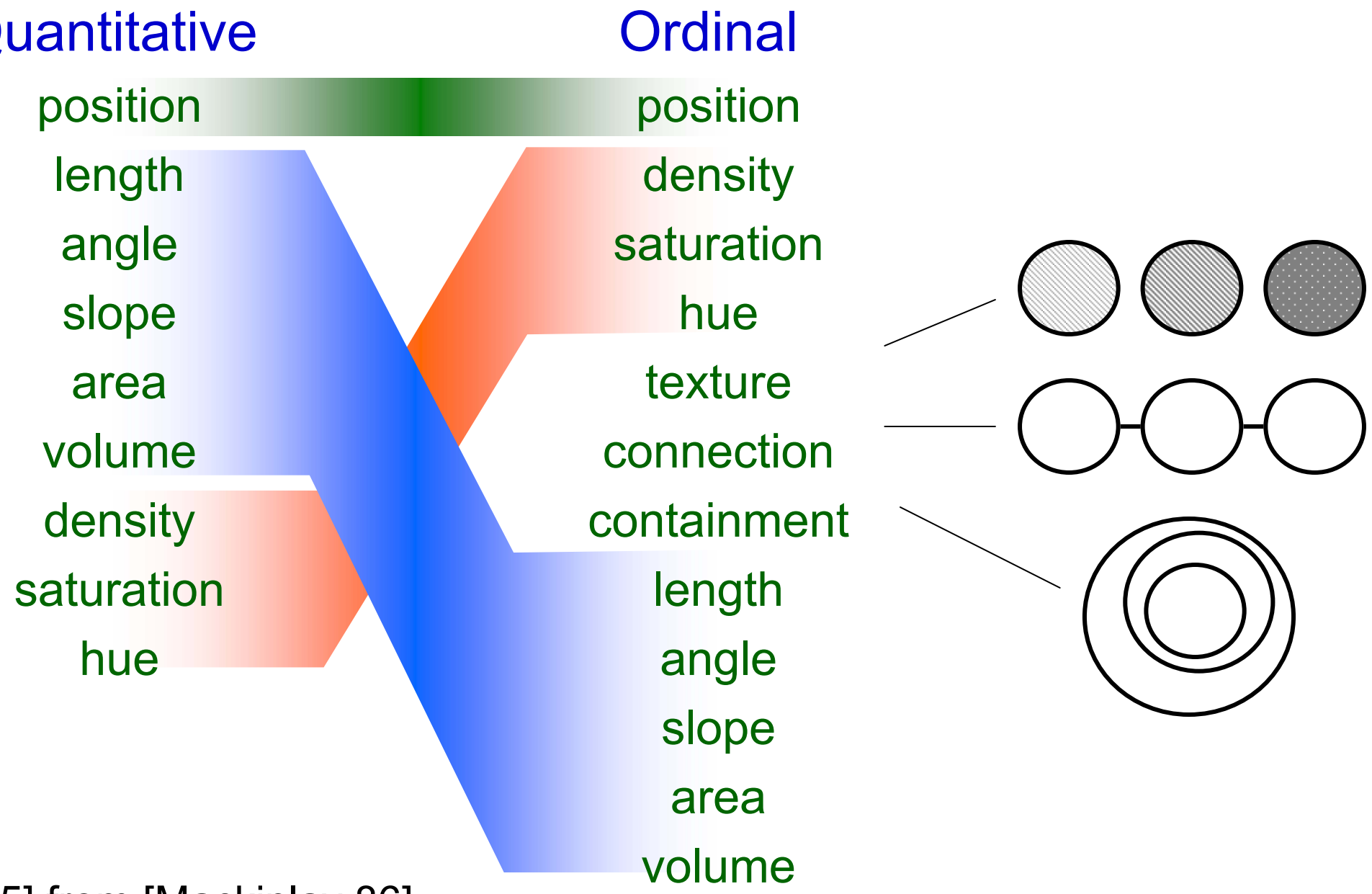
two dimensional value

5. Volume
6. Color/Density

three dimensional values

From [Hart 15]

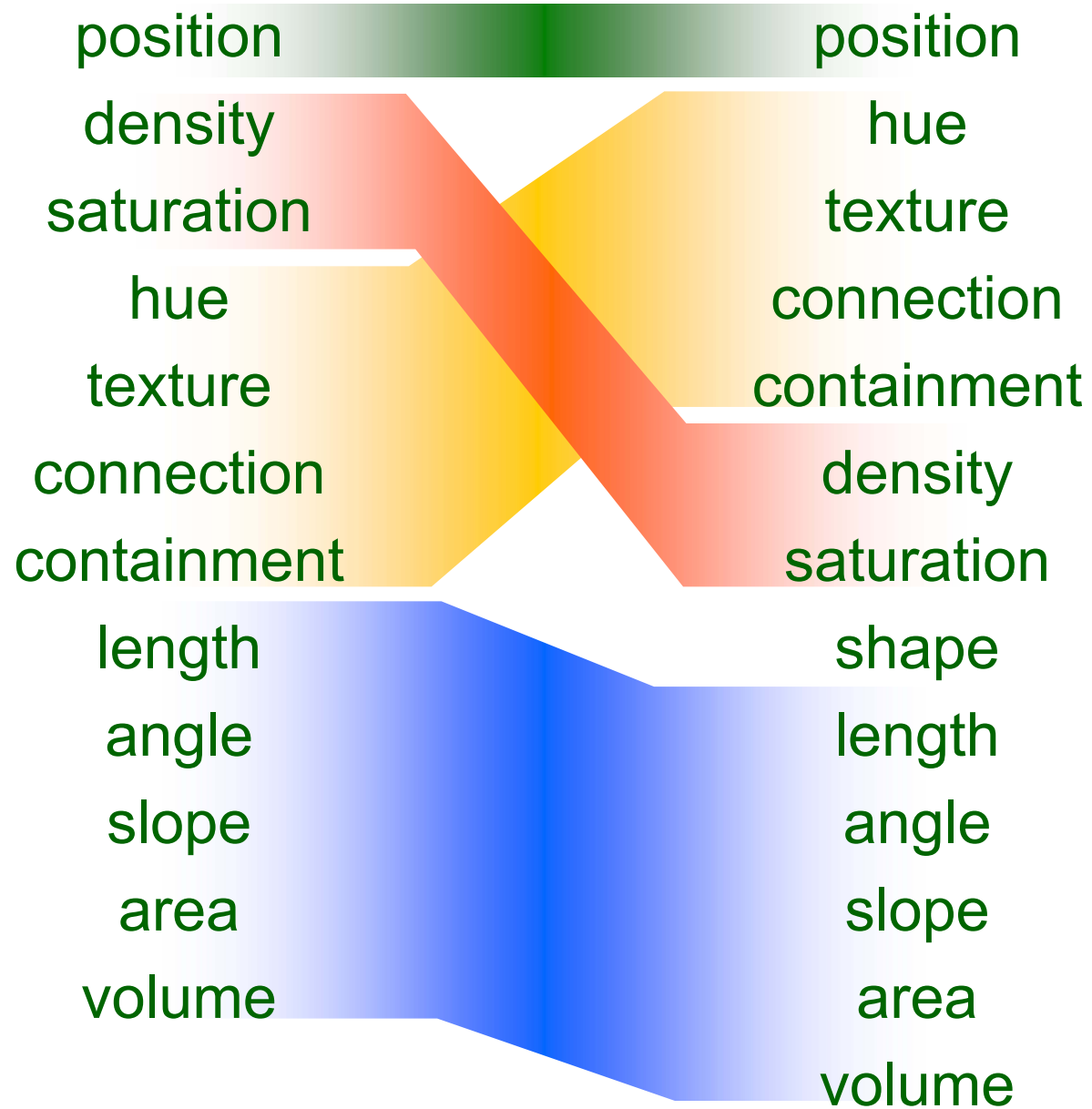
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[Hart 15] from [Mackinlay 86]

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Quantitative	Ordinal	Nominal
position	position	position
length	density	hue
angle	saturation	texture
slope	hue	connection
area	texture	containment
volume	connection	density
density	containment	saturation
saturation	length	shape
hue	angle	length
	slope	angle
	area	slope
	volume	area
		volume



[Hart 15] from [Mackinlay 86]

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Mapping guidelines

- Position is the best mapping for any kind of variables
  - but position is a limited resource
- For quantitative variables prefer continuous geometric features
  - length, angle, slope, area, volume
- For ordinal variables prefer density and saturation
- For nominal variables prefer hue, texture, and connection

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